

# Drift Separation and Certified Partial Refresh for RaBitQ under Streaming Inserts

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## Abstract

We analyze how a deployed RaBitQ index degrades when streaming inserts cause its centroid to drift away from the data it now serves, and study two endpoints of the deployment-policy spectrum: a *full-freeze* policy, which retains both the build-time routing centroids and the build-time stored data codes, and a *centroid-tracking* policy, which keeps routing centroids current online while leaving stored codes lazy. For the full-freeze policy under bounded centroid drift, we establish a *drift separation theorem* with three structurally separable channels: distance estimation within any fixed cell remains drift-invariant, recovering RaBitQ’s single-frame guarantee verbatim; the inner-product oracle gap between the frozen sketch and a counterfactual rebuilt sketch grows linearly in the drift radius and is monitored by a coverage statistic  $\Phi_T$ ; retrieval-level deviation enters only through assignment and routing drift across Voronoi cell boundaries, controlled by boundary-band measures, with a conditional recall-preservation corollary following under a margin assumption. For the centroid-tracking policy, we give an exact signed-error decomposition of the stale-code estimator into a static RaBitQ term and a deterministic representation-drift bias, and prove a *certified partial refresh* theorem: under a deterministic drift certificate, refreshing exactly the sign-code coordinates whose build-time rotated magnitude lies below the certificate recovers the full current sign code, with expected update cost  $O(\Lambda D)$  per affected vector under a Haar-random rotation. Read together, the two analyses describe the same drift event routed into structurally different channels, and expose an *asymmetric repair cost* as the structural design principle separating the two policies: the within-cell channel admits exact, low-cost certified repair, while the inter-cell channel does not.

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# 1 Introduction

**Setting.** RaBitQ provides a randomized data-oblivious quantization scheme for high-dimensional approximate nearest neighbor (ANN) search, with provable distance estimation guarantees under a fixed normalization geometry [7]. The construction normalizes a database vector against a reference centroid, encodes the resulting unit vector by a sign pattern in a randomly rotated frame, and estimates query similarity through a calibrated estimator. The guarantees are stated for a static dataset: the centroid used to normalize the data at index time is the same one used to interpret distances at query time.

In practice, vector databases are dynamic. Insertions and deletions shift the empirical centroid of the dataset over time, while the index itself remains frozen at build time. The raw database vectors and the rotation matrix do not change, but the geometry against which they were originally normalized does. This report studies how a frozen RaBitQ index degrades under such centroid drift.

**Drift affects two things, not one.** A frozen RaBitQ index encodes two pieces of geometric information that depend on the centroid: the *estimation frame* used to normalize data and queries (which determines what inner product the estimator targets), and the *partition / routing geometry* used by an inverted-file (IVF) layout (which determines which stored points are searched for a given query). When the centroid drifts, both of these become stale relative to the current dataset, but a system designer need not update them together. Maintaining current centroids online is cheap—it requires only a streaming update of a handful of  $D$ -dimensional vectors. Rewriting all stored data codes is expensive—it touches every vector in the index. The deployment policy is the choice of how much of the original geometry the system updates, and how lazily.

**Two natural endpoints.** We consider two endpoints of this design space. *Full-freeze* (FF): both the routing centroids and the stored data codes are frozen at build time. The deployed system is unchanged after indexing. This is the cheapest policy, but it is also the one in which drift accumulates most freely. *Centroid-tracking* (CT): the system maintains current cell centroids  $\mathbf{c}_k^t$  online and uses them for query routing and query normalization, while leaving the stored data codes untouched until an explicit refresh. Stored codes are still tied to the build-time centroids, but the routing decisions and query frame are made against the current geometry. Hybrid policies between these endpoints are possible and outside this report’s scope.

**The asymmetric anatomy of drift.** The two endpoints expose drift through structurally different channels.

Under FF, single-centroid distance estimation is drift-invariant: the deployed estimator operates entirely in the build-time frame, and the centroid-parametric distance identity closes exactly there, leaving only the standard RaBitQ stochastic error (see Theorem 4.4 in Part I). The retrieval-level risk under FF is therefore not a within-cell phenomenon; it is an inter-cell phenomenon. As centroids drift, points near the stale Voronoi boundaries change cell membership, and queries can be routed to the wrong cell. Repairing this requires identifying which stored points have flipped assignment. Unlike sign codes—where a coordinate-wise margin bound certifies which sign bits cannot have changed—cell assignment is a discrete arg min without an analogous continuity structure. There is no certificate-style shortcut that uses only stale information; determining the affected points requires inspecting all stored points and recomputing each arg min against the fresh centroids. Repair under FF therefore scales with the dataset size, not with the magnitude of drift.

Under CT, the routing geometry is continuously kept current at low cost, so the inter-cell channel does not accumulate drift in the same way. The price is paid within cells: the stored code is tied to the old normalized direction, but the query is normalized against the current centroid, so the deployed estimator has a deterministic bias whose magnitude is governed by the displacement  $\|\mathbf{o}_t - \mathbf{o}_0\|_2$  between the old and current normalized directions (see Theorem 9.1 in Part II). This within-cell channel admits a *certified partial refresh*: under a deterministic drift certificate, the only sign-code coordinates that can possibly have flipped are those whose old rotated magnitude is below the certificate. Refreshing exactly this set recovers the full current sign code, with expected cost  $O(\Lambda D)$  per affected point under a Haar-random rotation, where  $\Lambda = \gamma\sqrt{D}$  is the dimensionless drift parameter.

The two channels thus carry asymmetric repair costs. The inter-cell channel that FF concentrates drift into has no cheap exact repair; the within-cell channel that CT concentrates drift into has one. This asymmetry is the structural reason CT is a sensible alternative to FF: the design choice is not that CT is uniformly better, but that CT routes drift toward the channel that admits inexpensive exact repair.

**Contributions and structure.** This report consists of two parts, originally developed as separate analyses, presented here under a unified deployment-spectrum framing. Part I (Sections 3 to 7) analyzes the FF policy: it formalizes the centroid-drift threat model on an empirical measure of the post-insert dataset, establishes single-centroid drift invariance of the deployed distance estimator, develops a representation-staleness diagnostic  $\Phi_T$  for monitoring how far the frozen sketch has diverged from a fresh-rebuilt one, analyzes inter-cell assignment and routing drift through boundary-band measures, and combines these results into a drift separation theorem identifying the inter-cell channel as the unique deterministic- $\varepsilon$  retrieval-level mechanism. Part II (Sections 8 to 11) analyzes the CT policy: it formalizes stale-code estimation under a current-centroid query frame, gives an exact signed-error decomposition into a static RaBitQ term and a deterministic representation-drift term, and proves the certified partial refresh theorem together with its expected update-cost scaling. The two parts use independent notation, with the correspondence between symbols left implicit by the structural role of each quantity. Section 12 returns to the unified framing and discusses the asymmetric repair cost as the structural design principle suggested by the analysis.

## 2 Related Work

### 2.1 Vector Quantization for Approximate Nearest Neighbor Search

Vector quantization is widely used to compress database vectors for approximate nearest neighbor search. Product quantization (PQ) represents a vector by splitting it into subvectors and quantizing each part separately, yielding compact codes and fast table-based distance estimates [10]. Optimized product quantization improves this approach by learning a rotation before quantization [8], while anisotropic vector quantization modifies the distortion objective to better preserve retrieval-relevant inner products [9]. These methods are central to practical vector-search systems, including Faiss and GPU-based billion-scale search implementations [5, 11].

The strength of these methods is their practical efficiency. Their limitation, for our purposes, is that the resulting distance estimator is often difficult to analyze sharply: the codebook is learned or heuristic, and the quantized vector is typically used as a proxy for the original vector. RaBitQ takes a different route by using a randomized data-oblivious codebook and a calibrated estimator with theoretical guarantees [7]. Our work builds on this analyzable structure and asks how it behaves when the normalization geometry used to construct the stored code changes over time.

### 2.2 Randomized and Data-Oblivious Quantization

Randomized quantization methods replace learned codebook structure with random geometric structure, making the resulting estimators easier to analyze. Early examples include random hyperplane hashing for angular similarity [3] and  $p$ -stable locality-sensitive hashing for Euclidean distances [4]. More generally, randomized dimension reduction and approximate inner-product compression ask how many bits are needed to preserve geometric quantities such as distances and inner products [1].

RaBitQ fits this data-oblivious viewpoint. It randomly rotates a scaled hypercube codebook, quantizes a normalized vector by its rotated sign pattern, and uses a calibration scalar to obtain a controlled estimator [7]. Related random-rotation and low-bit schemes appear in Quantized Johnson–Lindenstrauss transforms, one-bit distributed mean estimation methods such as DRIVE and EDEN, and online quantization methods such as TurboQuant [14–17]. These works provide static compression or estimation guarantees for fixed representations.

Our question is different. We do not introduce a new static quantizer or improve RaBitQ’s static distortion bound. Instead, we study what happens when a code produced by a randomized quantizer is reused after the normalized representation it encoded has changed. The resulting issue is not only quantization error, but stale-target error: the estimator may remain calibrated for the old vector while being biased for the current one.

### 2.3 Dynamic ANN, Freshness, and Drift

Dynamic ANN systems maintain search quality as vectors are inserted, deleted, or replaced. Graph-based systems such as DiskANN and FreshDiskANN address this problem at the index level, focusing on update efficiency, search accuracy, and freshness in large-scale or streaming settings [12, 13]. This line of work asks how to keep the search structure current as the stored set changes.

We study a narrower quantization-level issue. Dataset updates can shift the centroid used to normalize vectors. Then a raw vector may be unchanged, but its normalized representation under the current centroid differs from the one used to build its stored code. Our question is not how to maintain an entire dynamic ANN index, but how a calibrated quantization estimator behaves when reused under this changed normalization geometry.

This also differs from the broader concept-drift literature, which studies prediction under nonstationary data distributions [2, 6]. Our drift is geometric: the reference point used for normalization changes, producing a mismatch between a stored code and the current representation it is meant to approximate. Thus the failure mode is not merely that the database evolves, but that a static quantization guarantee is applied outside the geometry for which it was constructed.

### 3 Threat Model and Notation

This chapter sets up the problem analyzed in the remainder of the report: how a deployed RaBitQ index degrades when streaming inserts cause its centroid to drift away from the data it now serves. Section 3.1 fixes notation and recalls the RaBitQ estimator; Section 3.2 defines the empirical measure capturing the post-insert index state and its true centroid; Section 3.3 introduces the drift parameter  $\varepsilon$  on which all later bounds depend; Section 3.4 contrasts the *deployed* estimator against a *fully fresh oracle* that rebuilds against the current centroid, isolating drift as the sole source of discrepancy.

#### 3.1 Notation and the RaBitQ Estimator

We follow the standard RaBitQ convention [7]: raw data and query vectors are first residualized and normalized against a centroid, and the estimator then approximates the inner product between the resulting unit vectors.

**Residualization.** For a centroid  $\mathbf{c} \in \mathbb{R}^D$ , a raw data vector  $\mathbf{o}_r$ , and a raw query vector  $\mathbf{q}_r$ ,

$$\begin{aligned}\rho(\mathbf{o}_r; \mathbf{c}) &:= \|\mathbf{o}_r - \mathbf{c}\|_2, & \mathbf{o}(\mathbf{o}_r; \mathbf{c}) &:= \frac{\mathbf{o}_r - \mathbf{c}}{\rho(\mathbf{o}_r; \mathbf{c})}, \\ \sigma(\mathbf{q}_r; \mathbf{c}) &:= \|\mathbf{q}_r - \mathbf{c}\|_2, & \mathbf{q}(\mathbf{q}_r; \mathbf{c}) &:= \frac{\mathbf{q}_r - \mathbf{c}}{\sigma(\mathbf{q}_r; \mathbf{c})}.\end{aligned}$$

The index is built once, at time  $t = 0$ , against an *index-time centroid*  $\mathbf{c}_0$ . We abbreviate

$$\mathbf{o}_0 := \mathbf{o}(\mathbf{o}_r; \mathbf{c}_0), \quad \mathbf{q}_0 := \mathbf{q}(\mathbf{q}_r; \mathbf{c}_0), \quad \rho := \rho(\mathbf{o}_r; \mathbf{c}_0), \quad \sigma_0 := \sigma(\mathbf{q}_r; \mathbf{c}_0).$$

**Quantization.** Let  $\mathbf{P} \in \mathbb{R}^{D \times D}$  be a Haar-random orthogonal matrix, drawn once at index construction and fixed thereafter. The RaBitQ codebook entry of  $\mathbf{o}_0$  is built from a sign pattern in the rotated frame:

$$\begin{aligned}\bar{\mathbf{x}}_b &:= \mathbf{1}[\mathbf{P}^{-1}\mathbf{o}_0 \geq \mathbf{0}] \in \{0, 1\}^D,^1 \\ \bar{\mathbf{x}} &:= \frac{2\bar{\mathbf{x}}_b - \mathbf{1}}{\sqrt{D}}, \\ \bar{\mathbf{o}} &:= \mathbf{P} \bar{\mathbf{x}} \in \mathbb{S}^{D-1}.\end{aligned}$$

The associated scale factor (Gao and Long [7], Lemma 3.1) is

$$\nu(\mathbf{o}_r) := \frac{1}{\sqrt{D}} \|\mathbf{P}^{-1}\mathbf{o}_0\|_1 = \langle \bar{\mathbf{o}}, \mathbf{o}_0 \rangle. \quad (1)$$

**Distance estimator.** The exact squared distance decomposes as

$$\|\mathbf{o}_r - \mathbf{q}_r\|_2^2 = \rho^2 + \sigma_0^2 - 2\rho\sigma_0 \langle \mathbf{o}_0, \mathbf{q}_0 \rangle, \quad (2)$$

so estimating the distance reduces to estimating the cosine  $\alpha := \langle \mathbf{o}_0, \mathbf{q}_0 \rangle$ . RaBitQ does so via

$$\hat{\alpha} := \frac{\langle \bar{\mathbf{o}}, \mathbf{q}_0 \rangle}{\nu(\mathbf{o}_r)}, \quad \widehat{\text{dist}}^2 := \rho^2 + \sigma_0^2 - 2\rho\sigma_0 \hat{\alpha}. \quad (3)$$

For both the stale frame  $\mathbf{c}_0$  and the fresh frame  $\mathbf{c}_T$  (introduced in Section 3.2), we write the squared-distance estimator as

$$\widehat{\text{dist}}_t^2 := \rho_t^2 + \sigma_t^2 - 2\rho_t\sigma_t \hat{\alpha}_t, \quad t \in \{0, T\}, \quad (4)$$

where  $\rho_0 \equiv \rho$  and  $\rho_T, \sigma_T$  are the fresh-frame residual norms.

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<sup>1</sup>Ties (exact zeros) in the indicator are resolved by the  $\geq 0$  convention. For any fixed nonzero vector, ties occur with probability zero under the Haar measure on  $\mathbf{P}$ .



### 3.2 Empirical Measure and the True Centroid at Time $T$

The index is built on  $N_0$  initial points, with index-time centroid

$$\mathbf{c}_0 := \frac{1}{N_0} \sum_{i=1}^{N_0} \mathbf{o}_{r,i}.$$

After  $T$  streaming inserts, the deployed index physically holds  $N_0 + T$  points. We model its current state by the *empirical measure*

$$\hat{\mathcal{D}}_T := \frac{1}{N_0 + T} \sum_{i=1}^{N_0+T} \delta_{\mathbf{o}_{r,i}}. \quad (5)$$

All retrieval and expectation metrics are evaluated against this realized measure rather than against any hypothetical generating distribution. The *true centroid at time  $T$*  is then

$$\mathbf{c}_T := \mathbb{E}_{\mathbf{o}_r \sim \hat{\mathcal{D}}_T}[\mathbf{o}_r] = \frac{1}{N_0 + T} \sum_{i=1}^{N_0+T} \mathbf{o}_{r,i}. \quad (6)$$

### 3.3 Endpoint-Only Drift and Frozen-State Invariants

**Drift assumption.** As inserts arrive,  $\mathbf{c}_0$  becomes mismatched with  $\mathbf{c}_T$ . We parameterize this mismatch by a single deterministic bound.

**Assumption 3.1** (Endpoint-only centroid drift). The index-time and current empirical centroids satisfy

$$\boxed{\|\mathbf{c}_T - \mathbf{c}_0\|_2 \leq \varepsilon} \quad (7)$$

The whole analysis proceeds as a function of this endpoint discrepancy alone. We do *not* establish any path-dependent or trajectory-level bound on the centroid’s evolution.

**What stays frozen.** Because the index is not rebuilt, two ingredients are mechanically locked to the  $\mathbf{c}_0$ -frame:

1. **The projection matrix  $\mathbf{P}$** , sampled once at index construction.
2. **The per-point stored tuple  $(\bar{\mathbf{x}}_b^{(i)}, \nu^{(i)}, \rho^{(i)})$** , computed against  $\mathbf{c}_0$  and never refreshed.

Consequently, the geometric meaning of these stored quantities decouples—gradually, and quantifiably—from the true geometry of  $\hat{\mathcal{D}}_T$ .

**Regularity domain.** We assume nondegeneracy:  $\mathbf{o}_r \notin \{\mathbf{c}_0, \mathbf{c}_T\}$  and  $\mathbf{q}_r \notin \{\mathbf{c}_0, \mathbf{c}_T\}$ . To keep all subsequent bounds well-posed, we further restrict to points and queries with

$$\rho > \varepsilon \quad \text{and} \quad \sigma_0 > \varepsilon. \quad (8)$$

This guarantees  $\|\mathbf{o}_r - \mathbf{c}_T\|_2 \geq \rho - \varepsilon > 0$  and  $\|\mathbf{q}_r - \mathbf{c}_T\|_2 \geq \sigma_0 - \varepsilon > 0$ , so the fresh-frame normalizations introduced below never blow up.

### 3.4 Deployed Estimator vs. Fully Fresh Oracle

To isolate the effect of drift, we compare the deployed system against a counterfactual one that is identical except for being rebuilt against the current centroid.

**Definition 3.2** (Deployed estimator). Because both the stored representation and the runtime query residualization are locked to  $\mathbf{c}_0$ , the deployed estimator operates entirely in the stale frame:

$$\hat{\alpha}_0(\mathbf{o}_r, \mathbf{q}_r) := \frac{\langle \bar{\mathbf{o}}_0, \mathbf{q}_0 \rangle}{\nu(\mathbf{o}_r)}. \quad (9)$$

**Definition 3.3** (Fully fresh oracle estimator). The counterfactual rebuilds in the fresh frame: it residualizes both data and query against  $\mathbf{c}_T$ , yielding  $\mathbf{o}_T$  and  $\mathbf{q}_T$ , and uses the corresponding fresh-frame scale

$$\nu_T(\mathbf{o}_r) := \frac{1}{\sqrt{D}} \|\mathbf{P}^{-1} \mathbf{o}_T\|_1 = \langle \bar{\mathbf{o}}_T, \mathbf{o}_T \rangle. \quad (10)$$

Its estimator is

$$\hat{\alpha}_T(\mathbf{o}_r, \mathbf{q}_r) := \frac{\langle \bar{\mathbf{o}}_T, \mathbf{q}_T \rangle}{\nu_T(\mathbf{o}_r)}. \quad (11)$$

*Remark 3.4* (Shared randomness). The oracle reuses *the same Haar-random rotation*  $\mathbf{P}$ : it is counterfactual only in its centroid frame, not in the RaBitQ random seed. Conditional on a fixed frame, RaBitQ’s Theorem 3.2 applies to the corresponding estimator with probability over  $\mathbf{P}$ . Pinning  $\mathbf{P}$  thus cleanly attributes any divergence between  $\hat{\alpha}_0$  and  $\hat{\alpha}_T$  to drift rather than to randomness.

**Dual frame discrepancies and the oracle gap.** Since data and query are normalized differently in the two systems, drift manifests through two perturbations:

$$\Delta \mathbf{o} := \mathbf{o}_0 - \mathbf{o}_T \quad (\text{data-side}), \quad \Delta \mathbf{q} := \mathbf{q}_0 - \mathbf{q}_T \quad (\text{query-side}). \quad (12)$$

The retrieval-relevant manifestation of drift is the *oracle gap*.

**Definition 3.5** (Oracle gap).

$$\Delta \hat{\alpha}(\mathbf{o}_r, \mathbf{q}_r) := \hat{\alpha}_0(\mathbf{o}_r, \mathbf{q}_r) - \hat{\alpha}_T(\mathbf{o}_r, \mathbf{q}_r). \quad (13)$$

Section 4 will (i) bound the dual perturbations  $\Delta \mathbf{o}$  and  $\Delta \mathbf{q}$  in terms of  $\varepsilon$  and the residual norms, and (ii) propagate these bounds into a deterministic control on  $|\Delta \hat{\alpha}(\mathbf{o}_r, \mathbf{q}_r)|$ .

## 4 Single-Centroid Estimator Drift

We work throughout on the regularity domain established in Section 3.3:  $\rho > \varepsilon$  and  $\sigma_0 > \varepsilon$ . Set  $\delta := \mathbf{c}_T - \mathbf{c}_0$ , with  $\|\delta\|_2 \leq \varepsilon$  by Theorem 3.1. We abbreviate the fresh-frame residual norms as

$$\rho_T := \rho(\mathbf{o}_r; \mathbf{c}_T), \quad \sigma_T := \sigma(\mathbf{q}_r; \mathbf{c}_T),$$

both strictly positive on the regularity domain. We also write

$$\nu_t := \langle \bar{\mathbf{o}}_t, \mathbf{o}_t \rangle = \frac{1}{\sqrt{D}} \|\mathbf{P}^{-1} \mathbf{o}_t\|_1, \quad t \in \{0, T\},$$

recovering  $\nu \equiv \nu_0$  from Section 3.

**Plan.** We bound the dual frame discrepancies  $\Delta \mathbf{o}, \Delta \mathbf{q}$  on the unit sphere (Section 4.1), use them to control the deterministic gap between the two centroid-frame inner products (Section 4.2), restate RaBitQ’s Theorem 3.2 in a frame-localized simplified form (Section 4.3), and combine these primitives into two estimator-level results: a *drift-invariance* statement for the deployed squared-distance estimator (Section 4.4.1) and a representation-staleness bound on the oracle  $\alpha$ -gap (Section 4.4.2).

### 4.1 Frame discrepancies on the unit sphere

**Lemma 4.1** (Dual frame-discrepancy bound). *On the regularity domain,*

$$\|\Delta \mathbf{o}\|_2 \leq \frac{2\varepsilon}{\rho}, \quad \|\Delta \mathbf{q}\|_2 \leq \frac{2\varepsilon}{\sigma_0}.$$

*Proof.* For the data side, set  $\mathbf{u} := \mathbf{o}_r - \mathbf{c}_0$  and  $\mathbf{v} := \mathbf{o}_r - \mathbf{c}_T = \mathbf{u} - \delta$ , so  $\|\mathbf{u}\|_2 = \rho$  and  $\|\mathbf{v}\|_2 = \rho_T$ . The identity

$$\frac{\mathbf{u}}{\rho} - \frac{\mathbf{v}}{\rho_T} = \frac{\mathbf{u} - \mathbf{v}}{\rho} + \frac{\rho_T - \rho}{\rho} \cdot \frac{\mathbf{v}}{\rho_T},$$

combined with  $\|\mathbf{v}/\rho_T\|_2 = 1$ ,  $\|\mathbf{u} - \mathbf{v}\|_2 = \|\delta\|_2$ , and the reverse triangle inequality  $|\rho_T - \rho| \leq \|\delta\|_2$ , yields  $\|\Delta \mathbf{o}\|_2 \leq 2\varepsilon/\rho$ . The query-side bound follows by the substitution  $(\mathbf{o}_r, \rho, \rho_T) \mapsto (\mathbf{q}_r, \sigma_0, \sigma_T)$ .  $\square$

A refinement tightens the leading constant from 2 to 1 (Section A); since the qualitative threshold analysis of Section 5 is insensitive to this factor, we use the constant-2 form throughout the body for its uniform validity over the regularity domain.

### 4.2 Centroid-frame inner-product discrepancy

**Corollary 4.2** (Centroid-frame inner-product discrepancy). *On the regularity domain,*

$$|\langle \mathbf{o}_0, \mathbf{q}_0 \rangle - \langle \mathbf{o}_T, \mathbf{q}_T \rangle| \leq 2\varepsilon \left( \frac{1}{\rho} + \frac{1}{\sigma_0} \right).$$

*Proof.* Telescope through the dual discrepancies:

$$\langle \mathbf{o}_0, \mathbf{q}_0 \rangle - \langle \mathbf{o}_T, \mathbf{q}_T \rangle = \langle \Delta \mathbf{o}, \mathbf{q}_0 \rangle + \langle \mathbf{o}_T, \Delta \mathbf{q} \rangle.$$

Apply Cauchy–Schwarz to each summand using  $\|\mathbf{q}_0\|_2 = \|\mathbf{o}_T\|_2 = 1$ , then Theorem 4.1.  $\square$

This bounds the gap between two parameterizations of the *same* raw squared distance, before any quantization enters. We will see in Section 4.4.1 that this gap, when paired with its conjugate norm-product change, cancels exactly at the distance-estimator level.

### 4.3 Per-frame RaBitQ guarantee

**Corollary 4.3** (Simplified RaBitQ estimator bound). *Fix a frame  $t \in \{0, T\}$ . By RaBitQ Theorem 3.2 applied at  $\mathbf{c}_t$ , for every  $\delta \in (0, 1)$ , with probability at least  $1 - \delta$  over the Haar draw of  $\mathbf{P}$ ,*

$$|\hat{\alpha}_t - \langle \mathbf{o}_t, \mathbf{q}_t \rangle| \leq \frac{C\sqrt{\log(1/\delta)}}{\nu_t\sqrt{D}},$$

for an absolute constant  $C > 0$ , where  $\sqrt{1 - \nu_t^2} \leq 1$  and the change of  $\sqrt{D-1}$  to  $\sqrt{D}$  have been absorbed into  $C$ , and the original tail  $2e^{-c_0\epsilon_0^2}$  has been re-expressed in  $\delta$ -form via  $\epsilon_0 = \Theta(\sqrt{\log(1/\delta)})$ .

The deployed estimator instantiates Theorem 4.3 at  $t = 0$ ; the fully fresh oracle instantiates it at  $t = T$ . Both share the Haar matrix  $\mathbf{P}$ .

### 4.4 Estimator-level consequences

The dual frame discrepancies (Sections 4.1 and 4.2) and the per-frame RaBitQ guarantee (Section 4.3) feed two distinct composite results: the deployed squared-distance estimator inherits RaBitQ’s single-frame guarantee verbatim (Section 4.4.1), while the oracle  $\alpha$ -gap admits a representation-staleness decomposition with an explicit  $\varepsilon$ -dependent term (Section 4.4.2).

#### 4.4.1 Deployed distance estimator: drift invariance

Define the deployed squared-distance estimator

$$\widehat{\text{dist}}_0^2(\mathbf{o}_r, \mathbf{q}_r) := \rho^2 + \sigma_0^2 - 2\rho\sigma_0\hat{\alpha}_0(\mathbf{o}_r, \mathbf{q}_r), \quad (14)$$

which the runtime computes from the stored tuple  $(\bar{\mathbf{x}}_b, \nu_0, \rho)$  and the runtime quantities  $(\sigma_0, \mathbf{q}_0)$ .

**Theorem 4.4** (Drift invariance of the deployed distance estimator). *Suppose  $\rho > 0$  and  $\sigma_0 > 0$ . For any  $\delta \in (0, 1)$ , with probability at least  $1 - \delta$  over the Haar draw of  $\mathbf{P}$ ,*

$$\left| \widehat{\text{dist}}_0^2(\mathbf{o}_r, \mathbf{q}_r) - \|\mathbf{o}_r - \mathbf{q}_r\|_2^2 \right| \leq \frac{2C\rho\sigma_0\sqrt{\log(1/\delta)}}{\nu_0\sqrt{D}}.$$

The bound contains no deterministic  $\varepsilon$ -dependent drift term: the deployed estimator inherits, in the stale frame, RaBitQ’s original single-frame squared-distance guarantee verbatim.

*Proof.* The centroid-parametric identity holds *exactly* at any fixed centroid; in particular, at  $\mathbf{c}_0$ ,

$$\|\mathbf{o}_r - \mathbf{q}_r\|_2^2 = \rho^2 + \sigma_0^2 - 2\rho\sigma_0\langle \mathbf{o}_0, \mathbf{q}_0 \rangle.$$

Subtract this from Equation (14):

$$\widehat{\text{dist}}_0^2 - \|\mathbf{o}_r - \mathbf{q}_r\|_2^2 = -2\rho\sigma_0(\hat{\alpha}_0 - \langle \mathbf{o}_0, \mathbf{q}_0 \rangle).$$

Apply Theorem 4.3 at  $t = 0$ . □

The mechanism is structural: the deployed system targets the raw distance via the *exact* centroid-parametric identity at  $\mathbf{c}_0$ , and the runtime computes  $\sigma_0$  from  $\mathbf{q}_r$  and the stored  $\mathbf{c}_0$  on the fly. Any deterministic frame discrepancy between the two parameterizations *cancels* in the squared-distance estimator: the gap that would open at the inner-product level is closed at the distance level by the matching change in the norm coefficients.

**Corollary 4.5** (Deployed-oracle distance gap). *Define analogously  $\widehat{\text{dist}}_T^2 := \rho_T^2 + \sigma_T^2 - 2\rho_T\sigma_T \hat{\alpha}_T$ . Then*

$$\widehat{\text{dist}}_0^2 - \widehat{\text{dist}}_T^2 = -2\rho\sigma_0(\hat{\alpha}_0 - \langle \mathbf{o}_0, \mathbf{q}_0 \rangle) + 2\rho_T\sigma_T(\hat{\alpha}_T - \langle \mathbf{o}_T, \mathbf{q}_T \rangle),$$

*a weighted difference of two RaBitQ stochastic errors with no standalone deterministic frame-discrepancy term.*

*Proof.* Apply the centroid-parametric identity at both  $\mathbf{c}_0$  and  $\mathbf{c}_T$ ; both right-hand sides equal  $\|\mathbf{o}_r - \mathbf{q}_r\|_2^2$  and cancel when differenced.  $\square$

#### 4.4.2 Oracle $\alpha$ -gap: representation staleness

**Theorem 4.6** (Oracle  $\alpha$ -gap decomposition). *Under Theorem 3.1 and the regularity domain, for any  $\delta \in (0, 1)$ , with probability at least  $1 - \delta$  over the Haar draw of  $\mathbf{P}$ ,*

$$|\Delta\hat{\alpha}(\mathbf{o}_r, \mathbf{q}_r)| \leq \underbrace{\frac{C\sqrt{\log(2/\delta)}}{\sqrt{D}}\left(\frac{1}{\nu_0} + \frac{1}{\nu_T}\right)}_{\mathcal{E}_{\text{quant}}} + \underbrace{2\varepsilon\left(\frac{1}{\rho} + \frac{1}{\sigma_0}\right)}_{\mathcal{E}_{\text{drift}}}.$$

*Proof.* Insert and subtract both centroid-frame inner products:

$$\Delta\hat{\alpha} = \underbrace{[\hat{\alpha}_0 - \langle \mathbf{o}_0, \mathbf{q}_0 \rangle]}_{(A_0)} + \underbrace{[\langle \mathbf{o}_0, \mathbf{q}_0 \rangle - \langle \mathbf{o}_T, \mathbf{q}_T \rangle]}_{(B)} - \underbrace{[\hat{\alpha}_T - \langle \mathbf{o}_T, \mathbf{q}_T \rangle]}_{(A_T)}.$$

Apply Theorem 4.3 at confidence  $\delta/2$  to each of  $(A_0)$  and  $(A_T)$ ; union-bound. Bound  $(B)$  deterministically by Theorem 4.2. Triangle inequality.  $\square$

The union step ignores the coupling between  $(A_0)$  and  $(A_T)$  from the shared Haar matrix; tighter analysis is discussed in Section B. Theorem 4.6 quantifies the divergence between the deployed sketch and a fresh-rebuilt sketch — a *representation-staleness certificate* that is meaningful as a system-aware monitoring signal but, by Theorem 4.4, does not itself translate into a degradation of the deployed system’s pointwise distance estimate.

### 4.5 Properties of the bounds

#### (P1) Source separation.

The stochastic events in Theorems 4.4 and 4.6 are over the Haar draw of  $\mathbf{P}$  alone; the deterministic frame discrepancies  $\Delta\mathbf{o}, \Delta\mathbf{q}$  depend only on the raw vectors and centroids. The decomposition in Theorem 4.6 is additive — the  $\mathcal{E}_{\text{quant}}$  event and the  $\mathcal{E}_{\text{drift}}$  deterministic bound have no shared randomness.

#### (P2) Non-uniform staleness.

$\mathcal{E}_{\text{drift}} \propto \rho^{-1} + \sigma_0^{-1}$ : centroid-proximal pairs incur the largest representation-staleness, centroid-distal pairs the smallest. This non-uniformity is invisible at the deployed distance level (Theorem 4.4) but reappears in the multi-centroid analysis where it interacts with cell-boundary geometry.

#### (P3) Validity domain.

Theorem 4.6, Theorem 4.1, and Theorem 4.2 require the regularity domain  $\{\rho > \varepsilon\} \cap \{\sigma_0 > \varepsilon\}$ . Outside it, the deterministic term must be replaced by the trivial constant 2 (the maximal gap of two inner products in  $[-1, 1]$ ). Theorem 4.4 requires only  $\rho > 0$  and  $\sigma_0 > 0$  and is therefore strictly stronger than the regularity domain would suggest.

**(P4) Recovery of vanilla RaBitQ at  $\varepsilon = 0$ .**

When  $\mathbf{c}_0 = \mathbf{c}_T$ ,  $\Delta \mathbf{o} \equiv \Delta \mathbf{q} \equiv \mathbf{0}$ , and the shared Haar matrix and shared tie-breaking convention force  $\bar{\mathbf{x}}_b^T = \bar{\mathbf{x}}_b$ ,  $\bar{\mathbf{o}}_T = \bar{\mathbf{o}}_0$ ,  $\nu_T = \nu_0$ , and hence  $\hat{\alpha}_T \equiv \hat{\alpha}_0$  identically. So  $\Delta \hat{\alpha} \equiv 0$ , and Theorems 4.4 and 4.6 reduce to RaBitQ’s original guarantee.

## 4.6 Separation of concerns

*Remark 4.7.* Theorems 4.4 and 4.6 measure two distinct things and have two distinct uses. Theorem 4.4 is the single-centroid retrieval statement: within a fixed single-cluster candidate set, centroid drift introduces no additional deterministic ranking bias beyond the original RaBitQ estimation noise. Theorem 4.6 is a representation-staleness certificate quantifying how far the frozen sketch has drifted from a fresh-rebuilt sketch — a system-aware monitoring signal but not a retrieval bound in the single-centroid regime. The retrieval-level mechanism by which drift degrades ANN search is therefore *not* a within-cluster phenomenon; it is an inter-cluster phenomenon addressed in Section 6. Section 5, accordingly, will use Theorem 4.6 rather than Theorem 4.4 to derive the staleness threshold  $\varepsilon^*$ : the threshold is a representation-rebuild signal, not a retrieval-accuracy signal, by design.

## 5 Representation-Staleness Threshold

Theorem 4.6 supplies a high-probability decomposition of the oracle  $\alpha$ -gap into a stochastic quantization envelope and a deterministic drift envelope. Theorem 4.4 establishes that the deployed squared-distance estimator is drift-invariant. Together, these results identify  $|\Delta\hat{\alpha}|$  as a *representation-staleness signal*: a divergence between the frozen sketch and a fresh-rebuilt sketch that does *not*, in the single-centroid regime, translate into degraded retrieval. This chapter formalizes the drift level at which this signal crosses the quantization envelope.

Throughout this chapter, we condition on a *fixed realized endpoint*  $\mathbf{c}_T$ , so that  $\nu_T$  is a fixed (data-dependent) quantity rather than a function of a free drift parameter.

### 5.1 Pointwise effective residual scale

**Definition 5.1** (Effective residual scale). For a raw data–query pair  $(\mathbf{o}_r, \mathbf{q}_r)$  on the regularity domain,

$$r_{\text{eff}}(\mathbf{o}_r, \mathbf{q}_r) := \frac{\rho \sigma_0}{\rho + \sigma_0} = \frac{1}{2} \text{HM}(\rho, \sigma_0), \quad (15)$$

the half-harmonic-mean of the stale-frame residual norms.

The shape  $r_{\text{eff}} \propto \rho\sigma_0/(\rho + \sigma_0)$  is dictated by the algebraic form of  $\mathcal{E}_{\text{drift}}$  in Theorem 4.6: it is the natural inverse of the deterministic drift coefficient. Centroid-proximal pairs have small effective scale.

### 5.2 Pointwise crossover

**Theorem 5.2** (Pointwise staleness crossover). Fix  $\delta \in (0, 1)$  and a realized endpoint  $\mathbf{c}_T$ . Freeze  $\nu_0, \nu_T$  at their realized values for the pair  $(\mathbf{o}_r, \mathbf{q}_r)$ . The unique solution to  $\mathcal{E}_{\text{drift}}(\varepsilon) = \mathcal{E}_{\text{quant}}(\delta)$  in the bound of Theorem 4.6 is

$$\boxed{\varepsilon^*(\mathbf{o}_r, \mathbf{q}_r; \delta) = \frac{C\sqrt{\log(2/\delta)}}{2\sqrt{D}} \cdot \frac{\frac{1}{\nu_0} + \frac{1}{\nu_T}}{\frac{1}{\rho} + \frac{1}{\sigma_0}}.} \quad (16)$$

The crossover is meaningful only inside the regularity domain — that is, when  $\varepsilon^* < \min(\rho, \sigma_0)$ . Outside, the refined drift envelope from Theorem 4.6 is no longer informative and must be replaced by the trivial bound 2; we therefore classify such points separately as regularity-boundary crossings rather than assigning them a meaningful pointwise crossover.

*Proof.* For fixed realized  $\nu_0, \nu_T$ , the equation is affine in  $\varepsilon$  and yields a unique solution. Substituting  $\mathcal{E}_{\text{drift}}(\varepsilon) = 2\varepsilon(1/\rho + 1/\sigma_0)$  and equating to  $\mathcal{E}_{\text{quant}} = (C\sqrt{\log(2/\delta)}/\sqrt{D})(1/\nu_0 + 1/\nu_T)$  gives the boxed expression.  $\square$

Conditioning on a realized endpoint is essential: as a function of  $\mathbf{c}_T$ , the denominator  $\nu_T$  is fixed; treating it as a generic free quantity would introduce a circularity, since  $\nu_T$  is itself a function of drift direction.

**Corollary 5.3** (Pointwise asymptotic order). On the high-probability event  $\{\nu_0, \nu_T \geq \nu^{\min}\}$  for

an absolute constant  $\nu^{\min} > 0$ ,<sup>2</sup>

$$\varepsilon^*(\mathbf{o}_r, \mathbf{q}_r; \delta) = \Theta\left(\frac{\sqrt{\log(1/\delta)}}{\sqrt{D}} \cdot r_{\text{eff}}(\mathbf{o}_r, \mathbf{q}_r)\right).$$

The statement is pointwise: it holds at a fixed  $(\mathbf{o}_r, \mathbf{q}_r)$  pair with high probability over  $\mathbf{P}$ .

*Proof.* Since  $\nu_0, \nu_T \in (0, 1]$ ,  $1/\nu_0 + 1/\nu_T \geq 2$  unconditionally, giving the lower bound  $\varepsilon^* \geq (C\sqrt{\log(2/\delta)}/\sqrt{D}) \cdot r_{\text{eff}}$ . On the event  $\{\nu_0, \nu_T \geq \nu^{\min}\}$ ,  $1/\nu_0 + 1/\nu_T \leq 2/\nu^{\min}$ , giving the matching upper bound.  $\square$

### 5.3 Aggregate staleness diagnostics

The pointwise crossover varies across the indexed set according to the heterogeneity of  $r_{\text{eff}}$  — the non-uniformity flagged in (P2) of Section 4.5. A single global threshold conceals this structure. We define instead an aggregate diagnostic on the empirical measure  $\widehat{\mathcal{D}}_T$ , with explicit bookkeeping for points outside the regularity domain.

**Definition 5.4** (Conservative staleness coverage). Fix a query  $\mathbf{q}_r$ , observed drift  $\varepsilon$ , and confidence  $\delta$ . The *conservative staleness coverage* is

$$\Phi_T(\varepsilon, \mathbf{q}_r; \delta) := \frac{1}{N_0 + T} \sum_{i=1}^{N_0+T} \mathbf{1}[\rho_i \leq \varepsilon \text{ or } \sigma_0 \leq \varepsilon \text{ or } \varepsilon^*(\mathbf{o}_{r,i}, \mathbf{q}_r; \delta) < \varepsilon]. \quad (17)$$

A point is counted as “stale” if it is either *outside the regularity domain* (the deterministic envelope has degenerated) or *inside the regularity domain with a crossed pointwise threshold*. Out-of-regularity points are precisely those where  $\mathcal{E}_{\text{drift}}$  activates the trivial-fallback constant 2, which dominates  $\mathcal{E}_{\text{quant}} = O(1/\sqrt{D})$  for any practical  $D$ .

The condition  $\sigma_0 \leq \varepsilon$  is a query-side regularity failure: when the query lies within  $\varepsilon$  of the stale centroid it activates uniformly across the index, driving  $\Phi_T(\varepsilon, \mathbf{q}_r; \delta) = 1$  regardless of any data-side state. This is mathematically consistent with the conservative construction — the query-side envelope has degenerated globally — and signals to the operator that the diagnostic is uninformative for that particular query.

**Domain-restricted alternative.** For diagnostic comparison one may also define

$$\Phi_T^{\text{reg}}(\varepsilon, \mathbf{q}_r; \delta) := \frac{\sum_i \mathbf{1}[\rho_i > \varepsilon, \sigma_0 > \varepsilon] \cdot \mathbf{1}[\varepsilon^*(\mathbf{o}_{r,i}, \mathbf{q}_r; \delta) < \varepsilon]}{\sum_i \mathbf{1}[\rho_i > \varepsilon, \sigma_0 > \varepsilon]}, \quad (18)$$

restricting to points where the deterministic envelope is informative. We treat  $\Phi_T$  as the primary signal because (i) it admits direct estimation without conditional denominators that themselves shift with  $\varepsilon$ , (ii) it generalizes verbatim to multi-cluster IVF, and (iii) it does not silently exclude points that have moved fastest under drift.  $\Phi_T^{\text{reg}}$  is useful for diagnosing whether crossings concentrate on out-of-regularity points (a separate failure mode) versus in-regularity ones.

**Definition 5.5** (Aggregate staleness quantile). Fix a query  $\mathbf{q}_r$ , confidence  $\delta$ , and tolerance  $q \in (0, 1)$ . The *q-staleness quantile* is the largest drift level at which conservative coverage stays below tolerance:

$$\varepsilon_q^*(\mathbf{q}_r; \delta) := \sup\{\varepsilon \geq 0 : \Phi_T(\varepsilon, \mathbf{q}_r; \delta) \leq q\}. \quad (19)$$

---

<sup>2</sup>RaBitQ Appendix B gives  $\mathbb{E}[\nu_t] \in (0.798, 0.800)$  for  $D \in [10^2, 10^6]$  with concentration  $\mathbb{P}(|\nu_t - \mathbb{E}[\nu_t]| > u/\sqrt{D}) \leq 2\exp(-cu^2)$ ; a pointwise lower bound  $\nu^{\min}$  follows directly. Aggregate use across  $N_0 + T$  indexed points (as in Theorem 5.6) requires a uniform lower bound, achievable via a union bound at a  $\log(N_0 + T)$  cost in the failure probability, or alternatively by instrumenting  $\nu_{0,i}, \nu_{T,i}$  directly from the index.



**Proposition 5.6** (Properties of  $\Phi_T$ ). *Under Theorem 3.1:*

1. Monotonicity in drift:  $\Phi_T(\varepsilon, \mathbf{q}_r; \delta)$  is non-decreasing in  $\varepsilon$ .
2. Monotonicity in confidence:  $\Phi_T(\varepsilon, \mathbf{q}_r; \delta)$  is non-decreasing in  $\delta$  (equivalently non-increasing in confidence  $1 - \delta$ ).
3. Recovery at  $\varepsilon = 0$ :  $\Phi_T(0, \mathbf{q}_r; \delta) = 0$  for all  $\delta < 1$ .
4. Aggregate dimensional scaling. *On the uniform  $\nu^{\min}$ -event (achievable across the  $N_0 + T$  indexed points via a union bound, paying  $\log(N_0 + T)$  in the failure probability), the pointwise threshold satisfies  $\varepsilon_i^* \asymp \sqrt{\log(1/\delta)/D} \cdot r_{\text{eff},i}$ . The conservative coverage's effective per-point crossing time is*

$$\tau_i := \min\{\rho_i, \sigma_0, \varepsilon_i^*\},$$

so  $\varepsilon_q^*$  is the empirical  $q$ -quantile of  $\{\tau_i\}$ . In the regime where the threshold branch binds (i.e.,  $\varepsilon_i^* < \min(\rho_i, \sigma_0)$  on the relevant quantile range), this reduces to

$$\varepsilon_q^*(\mathbf{q}_r; \delta) \asymp \sqrt{\frac{\log(1/\delta)}{D}} \cdot r_{\text{eff}}^{(q)},$$

where  $r_{\text{eff}}^{(q)}$  is the  $q$ -quantile of  $r_{\text{eff}}(\cdot, \mathbf{q}_r)$  under  $\hat{\mathcal{D}}_T$ .

*Proof.* (1) Both indicator components are monotone non-decreasing in  $\varepsilon$ . (2) From Theorem 5.2,  $\varepsilon^* \propto \sqrt{\log(2/\delta)}$  decreases as  $\delta$  increases, activating the threshold-crossing indicator earlier; the regularity-violation indicator is independent of  $\delta$ . (3) At  $\varepsilon = 0$ , regularity violation requires  $\rho_i \leq 0$  or  $\sigma_0 \leq 0$  (impossible by nondegeneracy of Section 3.3), and threshold crossing requires  $\varepsilon_i^* < 0$  (impossible since  $\varepsilon^* \geq 0$  for  $\delta < 1$ ). (4) follows from Theorem 5.3 applied uniformly over the  $\nu^{\min}$ -event, the definition of  $\tau_i$ , and the threshold-binding sub-regime; boundary measure-zero ambiguity is absorbed into absolute constants.  $\square$

## 5.4 Properties and operational reading

### (P1) Inverse-dimension scaling.

$\varepsilon^* \propto 1/\sqrt{D}$ . Higher-dimensional indices admit smaller drift before the staleness signal exceeds the quantization envelope — a counterintuitive consequence of RaBitQ's tighter quantization guarantee at large  $D$ .

### (P2) Confidence dependence.

$\varepsilon^* \propto \sqrt{\log(1/\delta)}$ . Stricter confidence enlarges the high-probability quantization envelope, so the crossover occurs at a larger drift magnitude. This is an envelope-tightness effect: stricter confidence makes the diagnostic more conservative, not the deployed system safer.

### (P3) Heterogeneity preservation.

Aggregating via  $\Phi_T$  preserves the per-point structure identified in (P2) of Section 4.5: centroid-proximal points cross first, distal points hold out longer. The aggregate signal is a coverage rate, not a homogenized average.

### (P4) Staleness, not retrieval.

Crossing  $\varepsilon^*$  does not imply degradation of any retrieval-accuracy quantity in the single-centroid regime (Theorem 4.4). The threshold flags representation divergence, not retrieval failure.

## 5.5 Remarks

*Remark 5.7* (What  $\varepsilon^*$  is and is not).  $\varepsilon^*$  is the drift level at which the deterministic representation-staleness envelope matches the high-probability RaBitQ quantization envelope. Below it, representation staleness is masked by quantization noise; above it, the deterministic envelope exceeds the quantization one. It is *not* a recall-degradation threshold, *not* a rebuild trigger (cost-benefit modeling is beyond this paper), and *not* a guarantee on deployed accuracy on any retrieval task. The drift-invariance of pointwise distance estimation (Theorem 4.4) holds at  $\varepsilon \gg \varepsilon^*$  exactly as at  $\varepsilon = 0$ . The retrieval consequences of drift, in the single-centroid regime, are absent; in multi-centroid IVF, they enter through assignment drift, not estimator staleness.

*Remark 5.8* (Why coverage rather than a global threshold). A scalar global threshold would average over the indexed set and obscure the structural non-uniformity in  $r_{\text{eff}}$ . Centroid-proximal points contribute disproportionately to staleness; an average smooths them away dishonestly. The conservative coverage formulation  $\Phi_T(\varepsilon)$  inverts this: it reports the share of the index that has crossed either the noise floor or the regularity boundary at the observed drift.

*Remark 5.9* (Drift-coupling of  $\nu_T$ ). The denominator  $\nu_T$  in Theorem 5.2 depends on  $\mathbf{c}_T$  and therefore on the realized drift. Conditioning on a fixed realized endpoint freezes  $\nu_T$  at its empirical value; Theorem 5.3’s  $\nu^{\min}$  assumption discharges the residual variability into a high-probability constant, at the cost of a slightly weaker bound. The constant-suppressed form is the relevant one for system-level monitoring; the explicit form of Theorem 5.2 is the one to instrument when finer per-point precision is needed.

*Remark 5.10* (Bridge to the IVF setting). The aggregate signal  $\Phi_T$  is defined so that it generalizes cleanly to multi-cluster IVF: replace  $\widehat{\mathcal{D}}_T$  by the empirical measure restricted to the probed cluster, and evaluate against the cluster’s own centroid drift  $\varepsilon^{(k)}$ . The retrieval-accuracy implication that the single-centroid setting does not support will, in IVF, route through cluster-assignment drift rather than through  $\Phi_T$  itself.  $\Phi_T$  remains a representation-staleness diagnostic at every level of the hierarchy.

## 6 Dual-Centroid Assignment Drift

Sections 4 and 5 settled the single-centroid story: pointwise distance estimation is drift-invariant (Theorem 4.4); the inner-product oracle gap  $|\Delta\hat{\alpha}|$  is a representation-staleness diagnostic (Theorems 4.6 and 5.2) but *not* a retrieval bound. This chapter enlarges the system to two centroids — the smallest setting in which a genuinely *retrieval-level* drift mechanism exists. Under multiple centroids, each data point and each query is *assigned* to one cluster by argmin distance to the centroid; if the centroids drift, the assignments may change. This phenomenon is absent in the single-centroid regime by construction; Theorem 6.7 establishes that it is the *unique deterministic- $\varepsilon$ -dependent retrieval mechanism* in the dual-centroid case.

### 6.1 Setup: stale and fresh centroids

The deployed system uses two stale centroids  $\mathbf{c}_1^0, \mathbf{c}_2^0 \in \mathbb{R}^D$  committed at index construction. After  $T$  streaming inserts, the empirical-mean centroids of the two clusters have drifted to  $\mathbf{c}_1^T, \mathbf{c}_2^T$ . We assume the *per-centroid* bounded-drift condition

$$\|\mathbf{c}_k^T - \mathbf{c}_k^0\|_2 \leq \varepsilon, \quad k \in \{1, 2\}, \quad (20)$$

generalizing Theorem 3.1 to the multi-centroid case, and that the centroids are well-separated:

$$\Delta_c := \|\mathbf{c}_2^0 - \mathbf{c}_1^0\|_2 > 0. \quad (21)$$

**Operational stale routing.** Generalizing the frozen-state condition of Section 3.3: at runtime, both insertion (cluster assignment for new data) and query routing rely entirely on the stored stale centroids; the deployed system never has access to  $\mathbf{c}_k^T$ .

The *stale* and *fresh* assignment functions are

$$a_0(\mathbf{x}) := \min_{k \in \{1, 2\}} \arg \min \|\mathbf{x} - \mathbf{c}_k^0\|_2^2, \quad a_T(\mathbf{x}) := \min_{k \in \{1, 2\}} \arg \min \|\mathbf{x} - \mathbf{c}_k^T\|_2^2, \quad (22)$$

where  $\min \arg \min$  resolves ties by smallest cluster index — shared across the stale and fresh systems for tie-breaking consistency. The *assignment-drift set* is

$$A_\Delta := \{\mathbf{x} \in \mathbb{R}^D : a_0(\mathbf{x}) \neq a_T(\mathbf{x})\}. \quad (23)$$

The stale and fresh *Voronoi boundaries* are the level sets of the *signed margin functions*

$$g_t(\mathbf{x}) := \|\mathbf{x} - \mathbf{c}_1^t\|_2^2 - \|\mathbf{x} - \mathbf{c}_2^t\|_2^2, \quad t \in \{0, T\}, \quad (24)$$

namely  $H_t := \{\mathbf{x} : g_t(\mathbf{x}) = 0\}$ . Since  $g_0(\mathbf{x}) = 2\langle \mathbf{x}, \mathbf{c}_2^0 - \mathbf{c}_1^0 \rangle + \|\mathbf{c}_1^0\|_2^2 - \|\mathbf{c}_2^0\|_2^2$  is affine in  $\mathbf{x}$ ,  $H_0$  is a hyperplane and

$$\text{dist}(\mathbf{x}, H_0) = \frac{|g_0(\mathbf{x})|}{2\Delta_c}. \quad (25)$$

With the shared deterministic tie-breaking convention,  $a_t(\mathbf{x}) = 1$  iff  $g_t(\mathbf{x}) \leq 0$ , so a flip in assignment forces  $g_0$  and  $g_T$  to lie in opposite (closed/open) half-lines about 0. In either case  $|g_T(\mathbf{x}) - g_0(\mathbf{x})| \geq |g_0(\mathbf{x})|$ , giving the implication

$$a_0(\mathbf{x}) \neq a_T(\mathbf{x}) \implies |g_0(\mathbf{x})| \leq |g_T(\mathbf{x}) - g_0(\mathbf{x})|. \quad (26)$$

## 6.2 Margin perturbation under centroid drift

Set  $\delta_k := \mathbf{c}_k^T - \mathbf{c}_k^0$ , with  $\|\delta_k\|_2 \leq \varepsilon$ .

**Lemma 6.1** (Margin perturbation). *For all  $\mathbf{x} \in \mathbb{R}^D$ ,*

$$|g_T(\mathbf{x}) - g_0(\mathbf{x})| \leq 2\varepsilon(\|\mathbf{x} - \mathbf{c}_1^0\|_2 + \|\mathbf{x} - \mathbf{c}_2^0\|_2) + \varepsilon^2.$$

*Proof.* Expand  $\|\mathbf{x} - \mathbf{c}_k^T\|_2^2 = \|\mathbf{x} - \mathbf{c}_k^0\|_2^2 - 2\langle \mathbf{x} - \mathbf{c}_k^0, \delta_k \rangle + \|\delta_k\|_2^2$ . Substituting into  $g_T - g_0$ :

$$g_T(\mathbf{x}) - g_0(\mathbf{x}) = -2\langle \mathbf{x} - \mathbf{c}_1^0, \delta_1 \rangle + 2\langle \mathbf{x} - \mathbf{c}_2^0, \delta_2 \rangle + \|\delta_1\|_2^2 - \|\delta_2\|_2^2.$$

Cauchy–Schwarz on the inner products and  $|\|\delta_1\|_2^2 - \|\delta_2\|_2^2| \leq \varepsilon^2$  give the stated bound.  $\square$

## 6.3 Boundary-band inclusion

**Corollary 6.2** (Data-dependent boundary band). *The assignment-drift set is contained in a data-dependent band around the stale boundary:*

$$A_\Delta \subseteq \left\{ \mathbf{x} : \text{dist}(\mathbf{x}, H_0) \leq \frac{\varepsilon(\|\mathbf{x} - \mathbf{c}_1^0\|_2 + \|\mathbf{x} - \mathbf{c}_2^0\|_2)}{\Delta_c} + \frac{\varepsilon^2}{2\Delta_c} \right\}.$$

*Proof.* If  $\mathbf{x} \in A_\Delta$ , Equation (26) gives  $|g_0(\mathbf{x})| \leq |g_T(\mathbf{x}) - g_0(\mathbf{x})|$ ; combine with Theorem 6.1 and  $\text{dist}(\mathbf{x}, H_0) = |g_0(\mathbf{x})|/(2\Delta_c)$ .  $\square$

The data-dependent form makes the geometry honest: the band is *narrower* near the centroids and *wider* far from them, in proportion to the residual norms. This anti-correlates with the effective scale  $r_{\text{eff}}$  of Theorem 5.1 and is the signature of the multi-centroid regime.

**Corollary 6.3** (Uniform-radius boundary band). *Suppose  $\|\mathbf{x} - \mathbf{c}_k^0\|_2 \leq R$  holds for  $k = 1, 2$  on the region of interest. Then*

$$A_\Delta \subseteq \mathcal{B}(H_0, \eta_R(\varepsilon)), \quad \eta_R(\varepsilon) := \frac{2R\varepsilon + \varepsilon^2/2}{\Delta_c},$$

where  $\mathcal{B}(H_0, \eta) := \{\mathbf{x} : \text{dist}(\mathbf{x}, H_0) \leq \eta\}$ . The radius  $R$  is a parameter of the local region; in Section 6.5 we instantiate it separately for the indexed and query supports.

In the small-drift regime  $\varepsilon \ll \Delta_c$ , the linear term dominates:  $\eta_R(\varepsilon) \approx 2R\varepsilon/\Delta_c$ .

## 6.4 Boundary-band measure

**Definition 6.4** (Boundary-band measure). For a probability measure  $\mu$  on  $\mathbb{R}^D$  and  $\eta > 0$ ,

$$M_\mu(H_0, \eta) := \mu(\mathcal{B}(H_0, \eta)) = \mu\{\mathbf{x} : \text{dist}(\mathbf{x}, H_0) \leq \eta\}.$$

**Proposition 6.5** (Assignment-drift bound). *Let  $\mu$  be any probability measure on  $\mathbb{R}^D$ . Under the assumptions of Theorem 6.3 with local radius  $R$  for the support of  $\mu$ ,*

$$\mu(A_\Delta) \leq M_\mu(H_0, \eta_R(\varepsilon)).$$

The data-dependent band of Theorem 6.2 yields the analogous bound

$$\mu(A_\Delta) \leq \mu\left\{ \mathbf{x} : \text{dist}(\mathbf{x}, H_0) \leq \frac{\varepsilon(\|\mathbf{x} - \mathbf{c}_1^0\|_2 + \|\mathbf{x} - \mathbf{c}_2^0\|_2)}{\Delta_c} + \frac{\varepsilon^2}{2\Delta_c} \right\},$$

which does not require a uniform radius bound.

*Proof.* Direct from the inclusions in Theorems 6.2 and 6.3.  $\square$

*Remark 6.6* (Linear scaling under boundary-density assumption). If  $\mu$  admits a density bounded above by  $\bar{f}$  in a neighborhood of  $H_0$ , and the slice  $H_0 \cap \text{supp}(\mu)$  has finite  $(D-1)$ -dimensional volume  $V_{D-1}$ , then  $M_\mu(H_0, \eta) \leq 2\bar{f} \cdot \eta \cdot V_{D-1}$ , giving  $\mu(A_\Delta) = O(\varepsilon)$  as  $\varepsilon \rightarrow 0$ . We do not impose this assumption: Theorem 6.5 holds in measure-theoretic generality. The density assumption recovers the linear-in- $\varepsilon$  rate that intuition suggests; the measure form is the form most useful for empirical instrumentation, where  $M_\mu$  can be estimated directly from the index without modeling the density or assuming finite slice volume.

## 6.5 Query-side routing drift

The same geometry applies to queries. Let  $\mathcal{Q}$  denote the query distribution on  $\mathbb{R}^D$ . The *routing-drift set* is

$$R_\Delta := \{\mathbf{q}_r \in \mathbb{R}^D : a_0(\mathbf{q}_r) \neq a_T(\mathbf{q}_r)\}. \quad (27)$$

Theorem 6.1, Theorems 6.2 and 6.3, and Theorem 6.5 apply verbatim with  $\mu \rightarrow \mathcal{Q}$  and  $\mathbf{x} \rightarrow \mathbf{q}_r$ . With a (possibly distinct) query-side local radius  $R_\mathcal{Q}$ ,

$$\mathcal{Q}(R_\Delta) \leq M_\mathcal{Q}(H_0, \eta_{R_\mathcal{Q}}(\varepsilon)). \quad (28)$$

The interpretation is asymmetric in operational consequence. Data-side  $\widehat{\mathcal{D}}_T(A_\Delta)$  counts indexed points whose stored cell membership is *miscategorized* relative to oracle (a property of the indexed dataset); query-side  $\mathcal{Q}(R_\Delta)$  counts queries that are *misrouted* (a property of the query distribution). Both contribute to candidate-set mismatch through distinct mechanisms.

## 6.6 Separability: candidate-set mismatch

We now assemble the central retrieval-level statement. For a query  $\mathbf{q}_r$ , let  $\mathcal{S}_0(\mathbf{q}_r)$  and  $\mathcal{S}_T(\mathbf{q}_r)$  denote the candidate sets returned by the deployed and fully fresh systems when probing the cluster routed-to under each system's centroids. We measure their disagreement by the symmetric-difference *mass* under the empirical indexed measure  $\widehat{\mathcal{D}}_T$ :

$$d_{\widehat{\mathcal{D}}_T}(\mathcal{S}_0, \mathcal{S}_T) := \widehat{\mathcal{D}}_T(\mathcal{S}_0 \triangle \mathcal{S}_T) \in [0, 1]. \quad (29)$$

**Theorem 6.7** (Candidate-set inclusion, by case). *For any query  $\mathbf{q}_r$ :*

1. If  $\mathbf{q}_r \notin R_\Delta$  (no routing drift), then both systems probe the same cluster  $k^* := a_0(\mathbf{q}_r) = a_T(\mathbf{q}_r)$ , and

$$\mathcal{S}_0(\mathbf{q}_r) \triangle \mathcal{S}_T(\mathbf{q}_r) \subseteq A_\Delta.$$

Within the common candidate set  $\mathcal{S}_0 \cap \mathcal{S}_T$ , the deployed and fresh squared-distance estimators differ only by RaBitQ stochastic noise:

$$\widehat{\text{dist}}_0^2 - \widehat{\text{dist}}_T^2 = \underbrace{(\widehat{\text{dist}}_0^2 - \|\mathbf{o}_r - \mathbf{q}_r\|_2^2)}_{\text{Theorem 4.4 at } \mathbf{c}_{k^*}^0} - \underbrace{(\widehat{\text{dist}}_T^2 - \|\mathbf{o}_r - \mathbf{q}_r\|_2^2)}_{\text{Theorem 4.4 at } \mathbf{c}_{k^*}^T},$$

with no deterministic  $\varepsilon$ -dependent term.

2. If  $\mathbf{q}_r \in R_\Delta$  (routing drift), then the two systems probe different clusters and  $\mathcal{S}_0(\mathbf{q}_r) \triangle \mathcal{S}_T(\mathbf{q}_r) \subseteq \text{index}$ .

*Proof.* Case 1: with both systems probing  $k^*$ , the deployed candidate set is  $\{\mathbf{o}_r \in \text{index} : a_0(\mathbf{o}_r) = k^*\}$  and the fresh set is  $\{\mathbf{o}_r \in \text{index} : a_T(\mathbf{o}_r) = k^*\}$ . A point lies in exactly one of these but not the other iff  $a_0(\mathbf{o}_r) \neq a_T(\mathbf{o}_r)$ , i.e.,  $\mathbf{o}_r \in A_\Delta$ . The within-common-set distance-invariance follows by Theorem 4.4 applied at each frame and triangle inequality. Case 2: trivial.  $\square$

**Proposition 6.8** (Expected candidate-set mismatch mass). *Under Equation (20) with per-centroid drift radius  $\varepsilon$ , and under the local-radius conditions  $R_\mu$  for the indexed support and  $R_Q$  for the query support,*

$$\begin{aligned} \mathbb{E}_{\mathbf{q}_r \sim Q}[d_{\widehat{\mathcal{D}}_T}(\mathcal{S}_0(\mathbf{q}_r), \mathcal{S}_T(\mathbf{q}_r))] &\leq Q(R_\Delta) + \widehat{\mathcal{D}}_T(A_\Delta) \\ &\leq M_Q(H_0, \eta_{R_Q}(\varepsilon)) + M_{\widehat{\mathcal{D}}_T}(H_0, \eta_{R_\mu}(\varepsilon)). \end{aligned}$$

*Proof.* For each  $\mathbf{q}_r$ , Theorem 6.7 gives

$$d_{\widehat{\mathcal{D}}_T}(\mathcal{S}_0, \mathcal{S}_T) \leq \mathbf{1}[\mathbf{q}_r \in R_\Delta] \cdot 1 + \mathbf{1}[\mathbf{q}_r \notin R_\Delta] \cdot \widehat{\mathcal{D}}_T(A_\Delta) \leq \mathbf{1}[\mathbf{q}_r \in R_\Delta] + \widehat{\mathcal{D}}_T(A_\Delta).$$

Take expectation over  $\mathbf{q}_r \sim Q$ ; apply Theorem 6.5 to each measure with its respective radius.  $\square$

The two contributions are *separable*: data-side assignment drift is a property of  $\widehat{\mathcal{D}}_T$  and the data-side residual norms; query-side routing drift is a property of  $Q$  and the query-side residual norms. Both are controlled by the *same* boundary band — a geometric quantity intrinsic to the centroid configuration. The expected-mass form is the right one because individual events  $\{\mathcal{S}_0 \neq \mathcal{S}_T\}$  are not bounded by drift mass: a single flipped indexed point can cause every routed query to see a candidate-set disagreement, even though that point has  $\widehat{\mathcal{D}}_T$ -mass  $1/(N_0 + T)$ .

The dual-centroid case is the minimal setting that exposes this mechanism. The  $K$ -centroid generalization is mechanically straightforward but conditionally heavy; we defer the formal statement to Section C.

## 6.7 Properties and limitations

### (P1) Separability of mechanisms.

Within-cell estimator behavior (Theorem 4.4;  $\Phi_T$  from Theorem 5.4) and inter-cell assignment/routing (Theorem 6.8) are mathematically separable. The first is a representation-monitoring quantity with no  $\varepsilon$ -dependent retrieval contribution; the second is the unique deterministic- $\varepsilon$ -dependent retrieval mechanism in the dual-centroid setting.

### (P2) Fragile-clustering regime.

When  $\Delta_c$  is small relative to  $\varepsilon$ , the boundary band  $\eta_R(\varepsilon) = O(\varepsilon/\Delta_c)$  becomes wide, and the upper bounds in Theorems 6.5 and 6.8 may exceed 1 (vacuous). This is honest: poorly separated centroids are precisely the configurations in which dual-centroid IVF is structurally fragile to drift, regardless of any analysis.

### (P3) Non-recall guarantee.

Theorem 6.8 bounds the expected mass of candidate-set disagreement, not the probability of recall failure. Recall depends additionally on the ranking within the candidate set, which Theorem 4.4 controls only up to RaBitQ stochastic noise; converting candidate-set agreement to ranking agreement requires an additional margin assumption (a gap between the  $k$ -th and  $(k + 1)$ -th true distances) introduced in Theorem 7.3.

### (P4) Independence from $\varepsilon^*$ .

The geometric scale  $\eta_R(\varepsilon)$  controlling assignment drift is unrelated to the representation-staleness threshold  $\varepsilon^*$  of Theorem 5.2. Both scale with  $\varepsilon$  but bound different objects:

$\varepsilon^*$  is when the inner-product staleness signal becomes statistically distinguishable from quantization noise;  $\eta_R(\varepsilon)$  is the boundary-band width within which assignment drift can occur. The two diagnostics are complementary.

## 6.8 Remarks

*Remark 6.9* (What this chapter establishes). In the dual-centroid setting, retrieval-level discrepancy between the deployed and fully fresh systems is fully attributable to assignment/routing drift. Within-cell, distance estimation remains drift-invariant. The retrieval contribution of drift is therefore a strictly inter-cell phenomenon, controlled by the boundary-band measure of the data and query distributions around the stale Voronoi hyperplane.

*Remark 6.10* (What this chapter does not establish). This chapter does not provide an event-probability bound for candidate-set inequality (only the expected mass of disagreement), does not provide a recall guarantee, does not re-enable an “ $\varepsilon^*$ -as-rebuild-trigger” interpretation, and does not analyze the dynamics of indexed points whose cell membership has flipped — those points are stored against the wrong centroid in the deployed system, and analysis of how their stale representation interacts with within-the-other-cell oracle representations is conceptually downstream.

*Remark 6.11* (Practical instrumentation). Both  $M_{\widehat{\mathcal{D}}_T}$  and  $M_Q$  are observable as approximations from the deployed system. A natural strategy: periodically estimate  $\mathbf{c}_k^T$  as the running empirical mean of recent inserts in cluster  $k$  under the *deployed* assignment  $a_0$ , then compute  $\eta_R(\varepsilon)$  from observed centroid drift and instrument  $M_{\widehat{\mathcal{D}}_T}(H_0, \eta_{R_\mu})$  by counting the fraction of indexed points within distance  $\eta_{R_\mu}$  of the stale boundary. The use of  $a_0$  rather than  $a_T$  for the running mean introduces an approximation error that vanishes in the small-drift regime where the assignments coincide on most points. Combined with the coverage signal  $\Phi_T$  of Theorem 5.4, this yields a two-axis drift dashboard:  $(\Phi_T, M_{\widehat{\mathcal{D}}_T}(H_0, \eta_{R_\mu}))$  — representation staleness on one axis, retrieval-relevant assignment drift on the other.

## 7 The Drift Separation Theorem

Sections 3 to 6 supplied a sequence of local results. This chapter collects them into one statement: *what does drift do to a frozen RaBitQ index, and where does each contribution live?* The answer is a separation theorem with three structurally distinct channels.

### 7.1 The three channels

Three distinct mechanisms govern the discrepancy between the deployed system and a fully fresh-rebuilt oracle. They are *channels* rather than additive components: each bounds a different operative object, and the structure of the result is one of accounting — naming what drift does, where, and to which quantity — rather than decomposing a single scalar error.

**Channel 1 — RaBitQ randomization noise.** The Haar-random rotation  $\mathbf{P}$  induces a high-probability quantization envelope on the RaBitQ estimator within any fixed centroid frame: by Theorem 4.3, with probability  $\geq 1 - \delta$ ,

$$|\hat{\alpha}_t - \langle \mathbf{o}_t, \mathbf{q}_t \rangle| \leq \frac{C\sqrt{\log(1/\delta)}}{\nu_t\sqrt{D}}, \quad t \in \{0, T\}.$$

This envelope has no  $\varepsilon$ -dependence; it is the original RaBitQ guarantee, applied frame-locally.

**Channel 2 — Within-cell representation staleness (single-centroid).** The deployed and fresh-rebuilt sketches diverge at the inner-product level by Theorem 4.6,

$$|\Delta\hat{\alpha}| \leq \mathcal{E}_{\text{quant}}(\delta) + \mathcal{E}_{\text{drift}}(\varepsilon),$$

where  $\mathcal{E}_{\text{drift}} \propto \varepsilon \cdot (1/\rho + 1/\sigma_0)$  is the only deterministic- $\varepsilon$  term in the channel. **Crucially, this term does not propagate to the squared-distance estimator:** Theorem 4.4 establishes that the deployed  $\widehat{\text{dist}}_0^2$  recovers the true raw squared distance with no  $\varepsilon$ -dependent bias, because the centroid-parametric identity at  $\mathbf{c}_0$  closes exactly. The Channel-2 deterministic term is therefore a *representation-divergence* signal, monitored by  $\Phi_T$  of Theorem 5.4, but *not* a retrieval-error contribution.

**Channel 3 — Inter-cell assignment/routing drift (multi-centroid).** When the index has  $K \geq 2$  clusters, drift induces a candidate-set discrepancy whose expected mass is bounded, by Theorem 6.8, by boundary-band measures:

$$\mathbb{E}_{\mathbf{q}_r \sim \mathcal{Q}}[d_{\widehat{\mathcal{D}}_T}(\mathcal{S}_0, \mathcal{S}_T)] \leq M_{\mathcal{Q}}(H_0, \eta_{R_{\mathcal{Q}}}(\varepsilon)) + M_{\widehat{\mathcal{D}}_T}(H_0, \eta_{R_{\mu}}(\varepsilon)).$$

This is the *unique deterministic- $\varepsilon$ -dependent retrieval-level term* in the entire accounting.

### 7.2 The separation theorem

The three channels are **structurally separable**: each is governed by a distinct object, with deterministic upper bounds that share no functional argument.

**Theorem 7.1** (Drift separation theorem; dual-centroid case). *Let the deployed RaBitQ system operate over the dual-centroid IVF index of Section 6, satisfying Theorem 3.1 with per-centroid drift radius  $\varepsilon$ . Fix confidence  $\delta \in (0, 1)$ . The discrepancy between the deployed system and a fully fresh-rebuilt oracle is governed by three structurally separable channels, summarized in Table 1.*

*The deterministic upper bounds at C2(b) and C3 share no functional argument:  $\mathcal{E}_{\text{drift}}$  depends on within-cell residual norms  $(\rho, \sigma_0)$  and the centroid drift, while  $M_{\mathcal{Q}} + M_{\widehat{\mathcal{D}}_T}$  depend on boundary-band*



|       | Bounded object                                                                                          | Bound                                                                          | Source of randomness                                                                          | $\varepsilon$ -dependence                                         |
|-------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| C1    | $ \widehat{\alpha}_t - \langle \mathbf{o}_t, \mathbf{q}_t \rangle $                                     | $O(1/(\nu_t \sqrt{D})), \text{ w.p. } 1 - \delta$                              | Haar $\mathbf{P}$                                                                             | None                                                              |
| C2(a) | $ \widehat{\text{dist}}_0^2 - \ \mathbf{o}_r - \mathbf{q}_r\ _2^2 $                                     | $O(\rho \sigma_0 / (\nu \sqrt{D})), \text{ w.p. } 1 - \delta$                  | Haar $\mathbf{P}$                                                                             | <b>None</b> (Theorem 4.4)                                         |
| C2(b) | $ \Delta \widehat{\alpha} $ (representation staleness)                                                  | $\mathcal{E}_{\text{quant}}(\delta) + \mathcal{E}_{\text{drift}}(\varepsilon)$ | Haar $\mathbf{P}$ (stoch.); $(\mathbf{c}_0, \mathbf{c}_T, \mathbf{o}_r, \mathbf{q}_r)$ (det.) | $O(\varepsilon \cdot r_{\text{eff}}^{-1})$ , <i>non-retrieval</i> |
| C3    | $\mathbb{E}_{\mathbf{q}_r \sim \mathcal{Q}}[d_{\widehat{\mathcal{D}}_T}(\mathcal{S}_0, \mathcal{S}_T)]$ | $M_{\mathcal{Q}} + M_{\widehat{\mathcal{D}}_T}$                                | $\widehat{\mathcal{D}}_T, \mathcal{Q}$ , centroid configuration                               | $O(\varepsilon \cdot R / \Delta_c)$ , <i>retrieval-level</i>      |

Table 1: The three channels of the drift separation theorem.

*geometry of the relevant measures. The randomness source at C3 — the empirical configurations of  $\widehat{\mathcal{D}}_T$  and  $\mathcal{Q}$  relative to the boundary  $H_0$  — is realized once and is independent of  $\mathbf{P}$ . Channel-by-channel guarantees combine into a joint guarantee across channels via a union bound, with no cross-term required.*

*Proof.* C1 is Theorem 4.3; C2(a) is Theorem 4.4; C2(b) is Theorem 4.6; C3 is Theorem 6.8. Structural separability is by inspection of the bounds: the deterministic terms at C2(b) and C3 are functions of disjoint geometric quantities, and the realized configurations of  $\widehat{\mathcal{D}}_T, \mathcal{Q}$  are independent of  $\mathbf{P}$ . Cross-channel union-bound combination is then standard.  $\square$

The structural content of the theorem is what each channel *does not* do:

- C1 and C2(a) are the within-cell story: they have *no* deterministic  $\varepsilon$ -dependence on the retrieval-relevant object (squared distance). The original RaBitQ guarantee survives drift verbatim, frame-by-frame.
- C2(b) has  $\varepsilon$ -dependence but on a *non-retrieval-relevant* object — the inner-product representation gap. It is a system-level monitoring quantity.
- C3 is the only channel where deterministic drift produces a retrieval-relevant deviation, and only via inter-cell mechanisms.

This separation is the central message: *drift in a frozen RaBitQ index is harmless within cells and dangerous only on cell boundaries.*

*Remark 7.2* (Extension to  $K > 2$  centroids). Theorem 7.1 is stated and proved formally for the dual-centroid case. The extension to  $K$ -centroid IVF replaces the C3 row with the pairwise-boundary union bound of Section C, under the conditions enumerated there. The structural-separability claim of Theorem 7.1 carries through unchanged; only the C3 expression becomes a sum.

### 7.3 Recall preservation under margin

Theorem 7.1 bounds candidate-set disagreement in expected mass, not recall. The standard bridge to recall requires an additional margin assumption.

For a query  $\mathbf{q}_r$  and confidence  $\delta \in (0, 1)$ , define the *uniform candidate-set envelope*

$$E_{\max}(\mathbf{q}_r; \delta) := \max_{\mathbf{o}_r \in \mathcal{S}_0(\mathbf{q}_r)} \frac{2C \rho(\mathbf{o}_r) \sigma_0 \sqrt{\log(|\mathcal{S}_0(\mathbf{q}_r)|/\delta)}}{\nu(\mathbf{o}_r) \sqrt{D}}, \quad (30)$$

the worst-case Theorem 4.4 envelope across all candidates in the deployed candidate set, with the  $\log(|\mathcal{S}_0|/\delta)$  factor coming from a union bound over candidates.

**Corollary 7.3** (Conditional recall preservation). *Fix a query  $\mathbf{q}_r$  and a target rank  $k$ . Let  $\Delta_k(\mathbf{q}_r)$  denote the true squared-distance gap between the  $k$ -th and  $(k+1)$ -th nearest neighbors among  $\mathcal{S}_0(\mathbf{q}_r)$ . Suppose:*

1. (Sufficient candidate-set agreement.) *The query is not misrouted ( $\mathbf{q}_r \notin R_\Delta$ ) and no indexed point in the routed cell lies in  $A_\Delta$ . (Sufficient — though not necessary — for  $\mathcal{S}_0(\mathbf{q}_r) = \mathcal{S}_T(\mathbf{q}_r)$ .)*
2. (Margin condition.) *The true  $k$ -NN distance gap exceeds twice the uniform envelope:*

$$\Delta_k(\mathbf{q}_r) > 2 E_{\max}(\mathbf{q}_r; \delta).$$

*Then, with probability at least  $1 - \delta$  over the Haar draw of  $\mathbf{P}$ , the deployed system’s top- $k$  ranking among  $\mathcal{S}_0(\mathbf{q}_r)$  matches the true top- $k$  within  $\mathcal{S}_0(\mathbf{q}_r)$ .*

*Proof.* Under condition (1), the deployed and fresh candidate sets coincide. Within the candidate set, Theorem 4.4 gives, with probability  $\geq 1 - \delta/|\mathcal{S}_0|$  pointwise,  $|\widehat{\text{dist}}_0^2(\mathbf{o}_r, \mathbf{q}_r) - \|\mathbf{o}_r - \mathbf{q}_r\|_2^2| \leq 2C\rho(\mathbf{o}_r)\sigma_0\sqrt{\log(|\mathcal{S}_0|/\delta)}/(\nu(\mathbf{o}_r)\sqrt{D})$ . A union bound over the candidate set yields  $|\widehat{\text{dist}}_0^2 - \|\cdot\|_2^2| \leq E_{\max}$  uniformly across  $\mathcal{S}_0(\mathbf{q}_r)$ , with probability  $\geq 1 - \delta$ . The true ordering between any two candidates cannot be inverted by their estimated distances when their estimation errors sum to less than the true gap; condition (2) supplies  $\Delta_k > 2E_{\max}$ , exactly this gap.  $\square$

The corollary is *conditional*: it does not assert that the conditioning event in (1) has high probability. By Theorem 6.8, the *expected mass* of candidate-set disagreement is bounded — but the *event probability* of agreement is not directly controlled. What Theorem 7.3 *does* deliver is the cleanest conditional statement: when the geometric condition (1) and the ranking condition (2) both hold, recall is preserved within the deployed candidate set. Practically, the operator monitors  $M_Q + M_{\widehat{\mathcal{D}}_T}$  to bound aggregate candidate-set drift; for individual queries with high-confidence retrieval requirements, the margin condition (2) is checkable post-hoc from the deployed system’s own distance estimates and acts as an internal acceptance test.

## 7.4 The contribution stated plainly

*In a frozen RaBitQ index under bounded centroid drift, distance estimation within any fixed cell is drift-invariant; retrieval-level deviation enters only through assignment and routing drift across Voronoi cell boundaries.*

## 7.5 Remarks

*Remark 7.4* (Diagnostic dashboard). The two deterministic- $\varepsilon$  quantities admit direct empirical instrumentation, yielding a two-axis dashboard:  $\Phi_T(\varepsilon, \mathbf{q}_r; \delta)$  for representation-staleness coverage (Theorem 5.4), and the boundary-band masses  $M_{\widehat{\mathcal{D}}_T}, M_Q$  for retrieval-relevant drift (Theorem 6.4).  $\Phi_T \uparrow$  with  $M$  stable signals a stale sketch with structurally insulated retrieval;  $M \uparrow$  signals that retrieval-relevant drift has increased — at risk of, not yet evidence of, recall degradation.

*Remark 7.5* (Why “separability” is structural, not rhetorical). The three channels are not separable merely as a presentation convenience. The deterministic upper bounds at C2(b) and C3 are functions of disjoint geometric quantities — within-cell residual norms versus boundary-band measures — and the realized configurations of  $\widehat{\mathcal{D}}_T$  and  $\mathcal{Q}$  relative to the boundary  $H_0$  are independent of the Haar draw of  $\mathbf{P}$  that drives Channel 1. The bounds therefore do not require a multiplicative interaction term in any composition; joint guarantees combine via union bound across channels.

*Remark 7.6* (Scope). The separation theorem is established formally for the dual-centroid case and via the pairwise-boundary extension of Section C for  $K$ -centroid IVF; recall preservation is conditional on a margin assumption (Theorem 7.3); and streaming centroid dynamics is outside scope.

## 8 Problem Setup

We now formalize the setting studied in this paper. The goal of this section is to separate three objects that are identical in the static RaBitQ analysis but become distinct in a dynamic setting:

- the raw database vector,
- its index-time normalized representation,
- and its current normalized representation after the dataset geometry has changed.

RaBitQ provides a theoretically controlled estimator when the quantized code, calibration scalar, and target inner product are all defined using the same normalized vector. Our setting keeps the same quantization mechanism, but allows the normalization geometry to change after indexing. This creates a mismatch between the representation encoded by the stored code and the representation relevant at query time.

Throughout, we work in ambient dimension  $D \geq 2$ . For a vector  $x \in \mathbb{R}^D$ , we write  $\|x\|_2$  for its Euclidean norm and  $\langle x, y \rangle$  for the standard inner product. The unit sphere is denoted by

$$\mathbb{S}^{D-1} := \{x \in \mathbb{R}^D : \|x\|_2 = 1\}.$$

The rest of this section proceeds in three steps. First, we recall the static RaBitQ notation needed for our analysis. Second, we introduce the centroid-induced representation drift model. Third, we define deterministic drift certificates, which will later be used to prove a certified partial refresh rule.

### 8.1 Static RaBitQ Notation

We recall the part of RaBitQ needed for our drift analysis. Let  $o, q \in \mathbb{S}^{D-1}$  be a normalized database vector and query vector. Define the scaled hypercube

$$\mathcal{C} = \left\{ -\frac{1}{\sqrt{D}}, +\frac{1}{\sqrt{D}} \right\}^D \subset \mathbb{S}^{D-1}.$$

Let  $P \in \mathbb{R}^{D \times D}$  be orthogonal; in RaBitQ,  $P$  is sampled from the Haar distribution on  $O(D)$  [7]. For a normalized vector  $o$ , write

$$y = P^\top o$$

for its rotated coordinates.

We use the deterministic tie-breaking sign convention

$$\text{sign}_+(u) := \begin{cases} +1, & u \geq 0, \\ -1, & u < 0. \end{cases}$$

The RaBitQ sign code is<sup>3</sup>

$$\sigma_P(o)_j := \text{sign}_+(y_j), \quad j = 1, \dots, D,$$

equivalently represented by the bit code  $b_P(o)_j = \mathbf{1}\{y_j \geq 0\}$ . The corresponding quantized vector is

$$Q_P(o) := P \frac{\sigma_P(o)}{\sqrt{D}}.$$

---

<sup>3</sup>The symbol  $\sigma$  in Part II (Sections 8–11) denotes the RaBitQ sign code, an element of  $\{-1, +1\}^D$ . This is unrelated to the query residual norm  $\sigma$  used in Part I (Section 3.1), which is a positive scalar. The two parts use independent notation; the shared Haar rotation  $P$  plays the same role throughout.

The sign rule is the nearest-codeword rule for the rotated hypercube. Indeed, for fixed  $P$  and  $o$ , minimizing  $\|o - Px\|_2$  over  $x \in \mathcal{C}$  is equivalent to maximizing  $\langle P^\top o, x \rangle$ , which is done coordinatewise by matching signs. We state this as Lemma 8.1, with proof in Appendix D.1.

**Lemma 8.1** (Nearest hypercube codeword). *Let  $P$  be orthogonal and let  $o \in \mathbb{S}^{D-1}$ . Then*

$$Q_P(o) = \arg \min_{v \in PC} \|o - v\|_2.$$

*Equivalently, if  $y = P^\top o$ , the minimizing codeword is obtained by choosing*

$$\sigma_j = \text{sign}_+(y_j), \quad Q_P(o) = P \frac{\sigma}{\sqrt{D}}.$$

The RaBitQ estimator also uses the alignment between the quantized vector and the normalized vector:

$$a(o) := \langle Q_P(o), o \rangle.$$

For query  $q \in \mathbb{S}^{D-1}$ , the static inner-product estimator is

$$\hat{s}_P(o, q) := \frac{\langle Q_P(o), q \rangle}{a(o)}.$$

We use the static RaBitQ guarantee as a black box: for fixed  $o, q$ , this estimator is unbiased for  $\langle o, q \rangle$  over the randomness of  $P$ , with the high-probability error control proved in [7]; we instantiate this bound in Corollary 9.6. Our analysis asks what changes when  $Q_P(o)$  and  $a(o)$  are reused after the relevant normalized vector has changed.

## 8.2 Representation Drift Model

We now define the source of staleness studied in this paper. Let  $\mathcal{X}^{(0)} \subset \mathbb{R}^D$  be the dataset at index time, with centroid  $c_0$ , and let  $\mathcal{X}^{(t)} \subset \mathbb{R}^D$  be the dataset at time  $t$ , with centroid  $c_t$ . For any point  $x \neq c$ , define the centroid-normalized direction

$$N_c(x) := \frac{x - c}{\|x - c\|_2} \in \mathbb{S}^{D-1}.$$

The normalized inner product determines centered squared distance once the radial terms are known. Indeed, if

$$o = N_c(x), \quad q = N_c(y),$$

and

$$\alpha = \|x - c\|_2, \quad \beta = \|y - c\|_2,$$

then

$$\|x - y\|_2^2 = \alpha^2 + \beta^2 - 2\alpha\beta \langle o, q \rangle.$$

Thus, in the centered RaBitQ setting, estimating squared distance reduces to estimating the normalized inner product. The proof is a direct expansion and is given in Appendix D.2.

Fix a raw database vector  $z \in \mathbb{R}^D$ . Its index-time normalized representation is

$$o_0 := N_{c_0}(z) = \frac{z - c_0}{\|z - c_0\|_2},$$

while its current normalized representation is

$$o_t := N_{c_t}(z) = \frac{z - c_t}{\|z - c_t\|_2}.$$

The raw vector  $z$  may be unchanged, but if  $c_t \neq c_0$ , then generally  $o_t \neq o_0$ . This centroid-induced change in normalized direction is the representation drift considered here.

A raw query  $q_r$  at time  $t$  is interpreted under the current centroid:

$$q_t := N_{c_t}(q_r) = \frac{q_r - c_t}{\|q_r - c_t\|_2}.$$

Therefore the current normalized target is

$$\langle o_t, q_t \rangle,$$

which determines the current centered squared distance through the identity above.

The stored RaBitQ code, however, was constructed from  $o_0$ . Let

$$\bar{o}_0 := Q_P(o_0), \quad a_0 := \langle \bar{o}_0, o_0 \rangle.$$

If the old code and old alignment scalar are reused at time  $t$ , the stale estimator is

$$\hat{s}_{P,t}^{\text{stale}} := \frac{\langle \bar{o}_0, q_t \rangle}{a_0}.$$

This estimator uses the current query direction  $q_t$ , but its database representation is still tied to  $o_0$ . The desired target is  $\langle o_t, q_t \rangle$ , so stale estimation is a mismatch between the old encoded direction and the current normalized direction.

*Remark 8.2* (Local centroids). We use a single centroid  $c_t$  for notation. In an inverted-file or clustered index, the same analysis applies locally by replacing  $c_t$  with the relevant cell centroid. The deterministic arguments only require two reference points,  $c_0$  and  $c_t$ .

### 8.3 Drift Certificates

We next record the deterministic information needed to reason about stale codes. A drift certificate is any upper bound on the change in normalized direction.

**Definition 8.3** (Drift certificate). For  $o_0 = N_{c_0}(z)$  and  $o_t = N_{c_t}(z)$ , a scalar  $\gamma_t \geq 0$  is a valid drift certificate at time  $t$  if

$$\|o_t - o_0\|_2 \leq \gamma_t.$$

The certificate is a deterministic promise, not a probabilistic model of drift. Once it holds, the decomposition and refresh guarantees below are deterministic. One sufficient certificate comes from bounding the centroid displacement

$$\Delta_t := \|c_t - c_0\|_2.$$

Since normalization is unstable near the centroid, we assume the vector is separated from both reference points.

**Assumption 8.4** (Normalization stability). There exists  $r_{\min} > 0$  such that

$$\|z - c_0\|_2 \geq r_{\min}, \quad \|z - c_t\|_2 \geq r_{\min}.$$

**Proposition 8.5** (Centroid displacement certificate). *Under Assumption 8.4,*

$$\|N_{c_t}(z) - N_{c_0}(z)\|_2 \leq \frac{2\|c_t - c_0\|_2}{r_{\min}}.$$

Hence

$$\gamma_t := \frac{2\Delta_t}{r_{\min}}$$

is a valid drift certificate.

*Proof.* See Appendix D.3. □

Orthogonality transfers the same certificate to the rotated coordinates. If

$$y_0 = P^\top o_0, \quad y_t = P^\top o_t,$$

then

$$\|y_t - y_0\|_2 = \|o_t - o_0\|_2 \leq \gamma_t.$$

Therefore, for every coordinate  $j$ ,

$$|(y_t)_j - (y_0)_j| \leq \gamma_t.$$

This coordinate-wise bound is the only input needed for certified partial refresh: any coordinate whose old rotated value is farther than  $\gamma_t$  from zero cannot change sign.

*Remark 8.6* (Global and vector-dependent certificates). Proposition 8.5 can be used with a global lower bound  $r_{\min}$  over many vectors, or with the sharper vector-dependent radius

$$r_t(z) := \min\{\|z - c_0\|_2, \|z - c_t\|_2\}.$$

The later deterministic arguments are unchanged; only the numerical value of the certificate changes.

## 9 Error Decomposition Under Drift

We now compare the stale estimator

$$\hat{s}_{P,t}^{\text{stale}} = \frac{\langle Q_P(o_0), q_t \rangle}{\langle Q_P(o_0), o_0 \rangle}$$

with the current target  $\langle o_t, q_t \rangle$ . The only difference from the static RaBitQ setting is that the stored code and alignment scalar were built from  $o_0$ , while the current centered distance is determined by  $o_t$ .

The signed error separates into two terms. One is the ordinary static RaBitQ error for the stale representation  $o_0$  and current query  $q_t$ . The other is a deterministic representation-drift term. Thus, stale estimation is not merely a larger quantization error: it estimates the wrong normalized target.

The results in this section are deterministic algebraic consequences of the definitions, except for the bias statement in Corollary 9.2 and the high-probability regime statement in Corollary 9.6, both of which invoke static RaBitQ guarantees from [7].

### 9.1 Stale RaBitQ Estimates the Wrong Target

The stale estimator reuses the old code and old alignment scalar:

$$\hat{s}_{P,t}^{\text{stale}} = \frac{\langle Q_P(o_0), q_t \rangle}{a_0}, \quad a_0 := \langle Q_P(o_0), o_0 \rangle.$$

The current target, however, is  $\langle o_t, q_t \rangle$ . Adding and subtracting  $\langle o_0, q_t \rangle$  gives the following exact identity.

**Proposition 9.1** (Exact stale-error decomposition). *Let  $o_0, o_t, q_t \in \mathbb{S}^{D-1}$ , and assume  $a_0 := \langle Q_P(o_0), o_0 \rangle \neq 0$ . Then*

$$\hat{s}_{P,t}^{\text{stale}} - \langle o_t, q_t \rangle = \left( \frac{\langle Q_P(o_0), q_t \rangle}{a_0} - \langle o_0, q_t \rangle \right) + \langle o_0 - o_t, q_t \rangle.$$

*Proof.* See Appendix E.1. □

The first term is exactly the static RaBitQ estimation error for the stale representation  $o_0$  and current query  $q_t$ . The second term is deterministic and depends only on the representation drift  $o_0 - o_t$  and the query direction. Thus the stale estimator is not merely a noisy estimate of the current target; it is centered around the old target  $\langle o_0, q_t \rangle$ .

### 9.2 Bias Induced by Representation Drift

In the static setting, RaBitQ is unbiased for the normalized vector used to construct the code. Applied to the fixed pair  $(o_0, q_t)$ , the static guarantee gives

$$\mathbb{E}_P \left[ \frac{\langle Q_P(o_0), q_t \rangle}{\langle Q_P(o_0), o_0 \rangle} \right] = \langle o_0, q_t \rangle.$$

But in the stale setting the current target is  $\langle o_t, q_t \rangle$ , not  $\langle o_0, q_t \rangle$ . Therefore the stale estimator can be unbiased for the old representation while biased for the current one.

**Corollary 9.2** (Bias of the stale estimator). *Assume the static RaBitQ unbiasedness guarantee applies to the fixed pair  $(o_0, q_t)$ . Then*

$$\mathbb{E}_P [\hat{s}_{P,t}^{\text{stale}}] - \langle o_t, q_t \rangle = \langle o_0 - o_t, q_t \rangle.$$



*Proof.* See Appendix E.2. □

The bias vanishes only when the drift is invisible to the query, for example when  $o_t = o_0$  or  $o_t - o_0 \perp q_t$ . Otherwise, the stale estimator is centered at the old target. This bias is not a consequence of random quantization error; it is caused by applying a static estimator to a changed normalized representation.

*Remark 9.3* (Direction, not calibration, must change). Consider any modified estimator of the form  $\langle Q_P(o_0), q_t \rangle / A$ , where  $A$  is any scalar computed from index-time data. Replacing  $a_0$  with such an  $A$  rescales only the first term in Proposition 9.1 and leaves the drift term  $\langle o_0 - o_t, q_t \rangle$  untouched. Therefore no scalar recalibration of the stored representation can remove the drift-induced bias; doing so requires updating the encoded direction itself, i.e., the sign code  $\sigma_P(o_0)$ . This motivates the partial-refresh rule developed in Section 10.

### 9.3 A Deterministic Drift Bound

The bias term from Corollary 9.2 depends on the query direction  $q_t$ . We now give a query-uniform bound: for any query direction, the drift contribution is controlled by the distance between the stale and current normalized representations.

**Lemma 9.4** (Drift inner-product bound). *For any  $o_0, o_t, q_t \in \mathbb{S}^{D-1}$ ,*

$$|\langle o_0 - o_t, q_t \rangle| \leq \|o_0 - o_t\|_2.$$

*Proof.* See Appendix E.3. □

This bound is deterministic and does not use any property of the random rotation  $P$ . It says that if the old and current normalized representations are close on the unit sphere, then the drift-induced bias is small for every query. Conversely, if the representation shift is large, then there may exist query directions for which the stale estimator has a large deterministic bias.

Combining this bound with the exact decomposition gives a total error bound that separates stochastic quantization error from deterministic representation drift.

**Corollary 9.5** (Combined stale-error bound). *Under the assumptions of Proposition 9.1,*

$$|\hat{s}_{P,t}^{\text{stale}} - \langle o_t, q_t \rangle| \leq \left| \frac{\langle Q_P(o_0), q_t \rangle}{a_0} - \langle o_0, q_t \rangle \right| + \|o_0 - o_t\|_2.$$

*Proof.* See Appendix E.4. □

The first term in Corollary 9.5 is exactly the static RaBitQ error for the stale representation  $o_0$  and current query  $q_t$ . The second term is independent of the quantization scheme; it measures the geometric mismatch between the representation used to build the code and the representation defining the current target.

If  $\gamma_t$  is a valid drift certificate, then  $\|o_t - o_0\|_2 \leq \gamma_t$ , so Corollary 9.5 gives

$$|\hat{s}_{P,t}^{\text{stale}} - \langle o_t, q_t \rangle| \leq \left| \frac{\langle Q_P(o_0), q_t \rangle}{a_0} - \langle o_0, q_t \rangle \right| + \gamma_t.$$

This is not a new theorem; it is the combined bound with the certificate substituted in.

## 9.4 When Drift Dominates Quantization

Corollary 9.5 separates the stale-estimator error into a stochastic quantization term and a deterministic drift term, but does not say which one dominates. To compare them, we instantiate the static RaBitQ high-probability guarantee from [7] for the fixed pair  $(o_0, q_t)$ . For every  $\epsilon_0 > 0$  there is an absolute constant  $c_0 > 0$  such that, with probability at least  $1 - 2e^{-c_0\epsilon_0^2}$  over  $P$ ,

$$\left| \frac{\langle Q_P(o_0), q_t \rangle}{a_0} - \langle o_0, q_t \rangle \right| \leq \sqrt{\frac{1 - a_0^2}{a_0^2}} \cdot \frac{\epsilon_0}{\sqrt{D - 1}}.$$

**Corollary 9.6** (High-probability stale-error bound). *Under Assumption 8.4, with probability at least  $1 - 2e^{-c_0\epsilon_0^2}$  over  $P$ ,*

$$|\hat{s}_{P,t}^{\text{stale}} - \langle o_t, q_t \rangle| \leq \sqrt{\frac{1 - a_0^2}{a_0^2}} \cdot \frac{\epsilon_0}{\sqrt{D - 1}} + \frac{2\Delta_t}{r_{\min}}.$$

*Proof.* Combine Corollary 9.5 with the static RaBitQ high-probability bound above and the centroid-displacement certificate of Proposition 8.5.  $\square$

The two terms have qualitatively different dependence on  $D$ . Equating them gives the critical drift scale

$$\Delta_t^{\text{crit}} := \frac{r_{\min}}{2} \sqrt{\frac{1 - a_0^2}{a_0^2}} \cdot \frac{\epsilon_0}{\sqrt{D - 1}} = \Theta\left(\frac{1}{\sqrt{D}}\right).$$

Here  $a_0$  is random through the Haar rotation, but in the static RaBitQ setting it concentrates around  $\sqrt{2/\pi}$  as  $D \rightarrow \infty$ . Thus its dependence on  $D$  does not change the  $\Theta(1/\sqrt{D})$  drift scale.

This separates two regimes:

- *Quantization-dominated* ( $\Delta_t \ll \Delta_t^{\text{crit}}$ ): the stochastic RaBitQ term controls the error and shrinks with  $D$ .
- *Drift-dominated* ( $\Delta_t \gg \Delta_t^{\text{crit}}$ ): the deterministic drift term controls the error and is independent of  $D$ .

A consequence is that increasing the ambient dimension improves the static quantization bound but *tightens* the centroid-drift tolerance  $\Delta_t^{\text{crit}}$ . In high dimensions, even small centroid shifts can dominate an otherwise sharply concentrated estimator. The certified refresh rule of Section 10 addresses precisely the drift-dominated regime.

## 10 Certified Partial Refresh

RaBitQ codes are sign patterns of randomly rotated normalized vectors. Let

$$y_0 := P^\top o_0, \quad y_t := P^\top o_t.$$

A full refresh would recompute the current sign vector

$$\text{sign}_+(y_t).$$

Certified partial refresh asks whether some coordinates can be safely left unchanged using only a deterministic drift certificate

$$\|o_t - o_0\|_2 \leq \gamma_t.$$

Since  $P$  is orthogonal, this certificate transfers exactly to the rotated coordinates:

$$\|y_t - y_0\|_2 = \|o_t - o_0\|_2 \leq \gamma_t.$$

Hence each coordinate changes by at most  $\gamma_t$ . If an index-time coordinate  $(y_0)_j$  is farther than  $\gamma_t$  from zero, then it cannot cross the sign boundary, so its sign is certified stable. Only coordinates with small index-time margin need to be recomputed.

The results in this section are deterministic. They use only the orthogonality of  $P$  and the validity of the drift certificate. Lemma 10.2 proves the coordinate-level stability test, and Theorem 10.3 proves that refreshing the uncertified coordinates recovers the same sign code as full re-quantization.

### 10.1 Margin Stability

Let

$$y_0 := P^\top o_0, \quad y_t := P^\top o_t.$$

A sign can change only if the corresponding rotated coordinate crosses zero. Thus the relevant stability quantity is the coordinate's distance from the sign boundary.

**Definition 10.1** (Coordinate margin). For  $j \in \{1, \dots, D\}$ , define the index-time margin

$$m_j := |(y_0)_j| = |(P^\top o_0)_j|.$$

If the coordinate perturbation is smaller than  $m_j$ , then coordinate  $j$  cannot cross zero. The drift certificate gives a uniform bound on all coordinate perturbations, yielding the following deterministic test.

**Lemma 10.2** (Certified sign stability). *Assume*

$$\|o_t - o_0\|_2 \leq \gamma_t.$$

*If*

$$|(y_0)_j| > \gamma_t,$$

*then*

$$\text{sign}_+((y_t)_j) = \text{sign}_+((y_0)_j).$$

*Proof.* See Appendix F.1. □

The lemma is one-sided. It certifies that high-margin coordinates cannot flip; it does not claim that low-margin coordinates must flip. Therefore the low-margin coordinates are not predicted flips, but the only coordinates whose signs are not certified stable. The partial refresh rule recomputes precisely these uncertified coordinates.

## 10.2 Certified Update Set

Lemma 10.2 certifies stability for coordinates with margin larger than  $\gamma_t$ . The only coordinates that remain uncertain are therefore

$$\mathcal{U}_t := \{j \in \{1, \dots, D\} : |(y_0)_j| \leq \gamma_t\}.$$

We call  $\mathcal{U}_t$  the certified update set. It is a conservative set: every possible sign flip is contained in  $\mathcal{U}_t$ , but not every coordinate in  $\mathcal{U}_t$  must actually flip.

**Theorem 10.3** (Certified partial refresh). *Assume*

$$\|o_t - o_0\|_2 \leq \gamma_t.$$

Let

$$\sigma_0 := \text{sign}_+(P^\top o_0), \quad \sigma_t := \text{sign}_+(P^\top o_t).$$

Define  $\tilde{\sigma} \in \{-1, +1\}^D$  by

$$\tilde{\sigma}_j = \begin{cases} \sigma_{t,j}, & j \in \mathcal{U}_t, \\ \sigma_{0,j}, & j \notin \mathcal{U}_t. \end{cases}$$

Then

$$\tilde{\sigma} = \sigma_t.$$

Consequently,

$$P \frac{\tilde{\sigma}}{\sqrt{D}} = Q_P(o_t).$$

*Proof.* See Appendix F.2. □

Thus refreshing only the coordinates in  $\mathcal{U}_t$  recovers exactly the sign code obtained by full re-quantization. The result is deterministic conditional on the drift certificate. Its conservativeness is only computational:  $\mathcal{U}_t$  may include coordinates whose signs do not actually change.

The theorem concerns the sign code. To match the full current RaBitQ estimator, one must also refresh current-vector-dependent scalar metadata. In particular, after recovering  $\sigma_t$ , the current quantized vector and alignment scalar are

$$Q_P(o_t) = P \frac{\sigma_t}{\sqrt{D}}, \quad a_t := \langle Q_P(o_t), o_t \rangle.$$

Therefore certified partial refresh gives the current code exactly, while the scalar  $a_t$  must be recomputed consistently for the refreshed representation.

## 10.3 Algorithmic Interpretation

The certified update set gives a simple maintenance rule. Store the old rotated coordinates

$$y_0 = P^\top o_0$$

and the old sign code

$$\sigma_0 = \text{sign}_+(y_0).$$

Given a valid drift certificate  $\gamma_t$ , form

$$\mathcal{U}_t = \{j : |(y_0)_j| \leq \gamma_t\}.$$

The algorithm keeps all signs outside  $\mathcal{U}_t$  and recomputes only the current signs inside  $\mathcal{U}_t$ .

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**Algorithm 1** Certified Partial Refresh

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**Require:** Old rotated coordinates  $y_0 = P^\top o_0$ , old sign code  $\sigma_0$ , drift certificate  $\gamma_t$ , access to  $(P^\top o_t)_j$  for  $j \in \mathcal{U}_t$

**Ensure:** Updated sign code  $\tilde{\sigma}$

- 1:  $\mathcal{U}_t \leftarrow \{j : |(y_0)_j| \leq \gamma_t\}$
  - 2:  $\tilde{\sigma} \leftarrow \sigma_0$
  - 3: **for**  $j \in \mathcal{U}_t$  **do**
  - 4:    $\tilde{\sigma}_j \leftarrow \text{sign}_+((P^\top o_t)_j)$
  - 5: **end for**
  - 6: **return**  $\tilde{\sigma}$
- 

By Theorem 10.3, if the certificate is valid, then

$$\tilde{\sigma} = \text{sign}_+(P^\top o_t).$$

Thus Algorithm 1 is not a heuristic bit update. It exactly reconstructs the full current sign code while touching only the uncertified coordinates.

The coordinate-level refresh cost is proportional to

$$|\mathcal{U}_t|.$$

When  $\gamma_t$  is small relative to the old margins, most signs are certified stable. When  $\gamma_t$  is large,  $\mathcal{U}_t$  can approach all of  $\{1, \dots, D\}$ , and the method degenerates to full re-quantization.

To evaluate the full current RaBitQ estimator, the current alignment scalar must also be refreshed:

$$a_t = \langle Q_P(o_t), o_t \rangle.$$

Therefore partial refresh exactly recovers the current sign code, while matching the full estimator also requires recomputing this scalar. Under the centroid-drift model, one valid certificate is

$$\gamma_t = \frac{2\|c_t - c_0\|_2}{r_{\min}},$$

provided Assumption 8.4 holds. Section 11 studies how large  $\mathcal{U}_t$  is expected to be under random rotation.

## 11 Scaling and Update Cost

The certified refresh rule is exact under a valid drift certificate, but its cost depends on the size of

$$\mathcal{U}_t = \{j : |(P^\top o_0)_j| \leq \gamma_t\}.$$

Small update sets give a cheap partial refresh; large update sets make the method approach full re-quantization.

This section studies the expected size of  $\mathcal{U}_t$  under the random-rotation model used by RaBitQ. Unlike the deterministic correctness result in Section 10, this cost analysis assumes

$$P \sim \text{Haar}(O(D)).$$

For fixed  $o_0 \in \mathbb{S}^{D-1}$ , this makes  $P^\top o_0$  uniformly distributed on the sphere. Therefore the expected update fraction is determined by the probability that one coordinate of a random sphere point has magnitude at most  $\gamma_t$ .

The relevant scale is not  $\gamma_t$  alone. A typical coordinate of a random unit vector in  $D$  dimensions has size  $1/\sqrt{D}$ , so the natural dimensionless drift parameter is

$$\Lambda_t := \gamma_t \sqrt{D}.$$

The results below make this scaling precise at the level of expectation and clarify what the analysis does, and does not, prove.

### 11.1 Random-Rotation Margin Distribution

We first record the margin distribution induced by the Haar rotation. If  $P \sim \text{Haar}(O(D))$  and  $o_0 \in \mathbb{S}^{D-1}$  is fixed, then

$$y_0 := P^\top o_0$$

is uniformly distributed on  $\mathbb{S}^{D-1}$ . Hence each margin

$$m_j := |(y_0)_j|$$

has the same marginal law as  $|X_1|$ , where  $X \sim \text{Unif}(\mathbb{S}^{D-1})$ .

**Proposition 11.1** (Coordinate distribution under random rotation). *Let  $X = (X_1, \dots, X_D) \sim \text{Unif}(\mathbb{S}^{D-1})$ . Then  $X_1$  has density*

$$p_D(u) = \frac{\Gamma(D/2)}{\sqrt{\pi} \Gamma((D-1)/2)} (1 - u^2)^{(D-3)/2}, \quad u \in [-1, 1].$$

Therefore, for  $0 \leq \gamma \leq 1$ ,

$$\mathbb{P}(|X_1| \leq \gamma) = 2 \int_0^\gamma \frac{\Gamma(D/2)}{\sqrt{\pi} \Gamma((D-1)/2)} (1 - u^2)^{(D-3)/2} du.$$

*Proof.* See Appendix G.2. □

This density is the basic object controlling update cost. Since the update set is

$$\mathcal{U}_t = \{j : |(y_0)_j| \leq \gamma_t\},$$

the expected update fraction is the probability that a random coordinate margin is at most  $\gamma_t$ . The full expectation identity is stated in Proposition 11.3. No coordinate independence is needed there; symmetry and linearity of expectation are enough.

For intuition, a coordinate of a random unit vector has scale  $1/\sqrt{D}$ . Equivalently,  $\sqrt{D}X_1$  is approximately standard normal in high dimension. Thus the update fraction is governed by the normalized threshold  $\gamma_t \sqrt{D}$ , which motivates the dimensionless drift parameter introduced next.

## 11.2 Dimensionless Drift Parameter

The update threshold  $\gamma_t$  should be compared to the typical coordinate scale of  $P^\top o_0$ . Under Haar rotation, this scale is  $1/\sqrt{D}$ , so we define

$$\Lambda_t := \gamma_t \sqrt{D}.$$

**Definition 11.2** (Dimensionless drift parameter). Given a drift certificate  $\gamma_t$ , the dimensionless drift parameter is

$$\Lambda_t := \gamma_t \sqrt{D}.$$

Thus  $\Lambda_t \ll 1$  means the certificate is small relative to a typical rotated coordinate margin, while  $\Lambda_t = \Theta(1)$  means the certificate is comparable to that margin.

**Proposition 11.3** (Expected certified update fraction). *Let  $P \sim \text{Haar}(O(D))$ , fix  $o_0 \in \mathbb{S}^{D-1}$ , and define*

$$y_0 = P^\top o_0, \quad \mathcal{U}_t = \{j : |(y_0)_j| \leq \gamma_t\}.$$

*Then*

$$\mathbb{E} \left[ \frac{|\mathcal{U}_t|}{D} \right] = \mathbb{P}(|X_1| \leq \min\{\gamma_t, 1\}), \quad X \sim \text{Unif}(\mathbb{S}^{D-1}).$$

*In particular, if  $0 \leq \gamma_t \leq 1$ , then*

$$\mathbb{E} \left[ \frac{|\mathcal{U}_t|}{D} \right] = 2 \int_0^{\gamma_t} \frac{\Gamma(D/2)}{\sqrt{\pi} \Gamma((D-1)/2)} (1-u^2)^{(D-3)/2} du.$$

*If  $\gamma_t \geq 1$ , then*

$$\mathbb{E} \left[ \frac{|\mathcal{U}_t|}{D} \right] = 1.$$

*Proof.* See Appendix G.3. □

**Corollary 11.4** (Asymptotic update fraction). *Fix  $\Lambda > 0$ , and set  $\gamma_t = \Lambda/\sqrt{D}$ , so that  $\Lambda_t = \gamma_t \sqrt{D} = \Lambda$ . Then, as  $D \rightarrow \infty$ ,*

$$\mathbb{E} \left[ \frac{|\mathcal{U}_t|}{D} \right] \rightarrow 2\Phi(\Lambda) - 1,$$

*where  $\Phi$  is the standard normal distribution function. In the small- $\Lambda$  regime,*

$$2\Phi(\Lambda) - 1 = \sqrt{\frac{2}{\pi}} \Lambda + O(\Lambda^3),$$

*and therefore*

$$\mathbb{E}[|\mathcal{U}_t|] \approx \sqrt{\frac{2}{\pi}} \Lambda_t D = \sqrt{\frac{2}{\pi}} \gamma_t D^{3/2}.$$

*Proof.* By Proposition 11.3, when  $\gamma_t = \Lambda/\sqrt{D}$ ,

$$\mathbb{E} \left[ \frac{|\mathcal{U}_t|}{D} \right] = \mathbb{P}(\sqrt{D} |X_1| \leq \Lambda), \quad X \sim \text{Unif}(\mathbb{S}^{D-1}).$$

A standard marginal limit for a uniformly random sphere point gives

$$\sqrt{D} X_1 \Rightarrow Z, \quad Z \sim \mathcal{N}(0, 1),$$

so the probability converges to  $\mathbb{P}(|Z| \leq \Lambda) = 2\Phi(\Lambda) - 1$ . The Taylor expansion of  $\Phi$  at zero gives the small- $\Lambda$  expression. □

Corollary 11.4 gives three regimes:

- If  $\Lambda_t \ll 1$ , only a small fraction of coordinates are uncertified.
- If  $\Lambda_t = \Theta(1)$ , certified refresh updates a constant-order fraction of the code.
- If  $\Lambda_t \gg 1$ , most coordinates may be uncertified, so the method approaches full re-quantization.

### 11.3 What This Section Can and Cannot Prove

The correctness theorem in Section 10 is deterministic: if a valid certificate satisfies

$$\|o_t - o_0\|_2 \leq \gamma_t,$$

then refreshing  $\mathcal{U}_t$  recovers the same sign code as full re-quantization. This does not require  $P$  to be random, only orthogonal.

The cost analysis in this section is different. To estimate

$$|\mathcal{U}_t| = |\{j : |(P^\top o_0)_j| \leq \gamma_t\}|,$$

we assume  $P \sim \text{Haar}(O(D))$ . Under that assumption,  $P^\top o_0$  is uniform on the sphere, giving an expected update fraction governed by  $\Lambda_t = \gamma_t \sqrt{D}$ .

This expectation is not a high-probability guarantee for every realized rotation. It also measures a conservative update set, not the actual number of sign flips: a coordinate is included in  $\mathcal{U}_t$  whenever it could flip under any drift of size  $\gamma_t$ , even if it does not flip for the realized drift. Finally, the theory assumes a valid drift certificate. Proposition 8.5 gives one sufficient certificate for centroid-induced drift, but other drift models would need their own certificates.

Thus the section proves expected update-cost scaling under Haar rotation, while exact refresh correctness remains the deterministic statement from Section 10.



## 12 Discussion: Asymmetric Repair Cost

Part I (Sections 3 to 7) analyzed centroid drift under the full-freeze (FF) policy, in which the deployed system retains both the build-time routing centroids and the build-time stored data codes. Part II (Sections 8 to 11) analyzed centroid drift under the centroid-tracking (CT) policy, in which the system maintains the current cell centroids online and uses them for routing and query normalization, while leaving stored data codes lazy. The two parts use different notational conventions and were developed as distinct analyses; this section returns to the unified deployment-spectrum framing introduced in Section 1 and reads the two analyses against each other.

### 12.1 What the two policies expose

The drift separation theorem (Theorem 7.1) shows that, under FF, the deployed pointwise distance estimator is drift-invariant within any fixed cell (Theorem 4.4); the only deterministic- $\varepsilon$  retrieval-level deviation enters through inter-cell assignment and routing drift, controlled by the boundary-band measures of Theorem 6.8. Within-cell representation staleness, captured by the oracle  $\alpha$ -gap of Theorem 4.6 and monitored by the coverage statistic  $\Phi_T$  of Theorem 5.4, is a representation-divergence signal but not a retrieval-error contribution.

Under CT, the inter-cell channel is structurally suppressed: the routing geometry and the query frame are computed against the current centroids, so cell assignment and routing decisions are made in fresh geometry. The stored data codes, however, remain tied to the build-time normalized directions. The deployed estimator therefore inherits the deterministic representation-drift term of Theorem 9.1,

$$\hat{s}_{P,t}^{\text{stale}} - \langle o_t, q_t \rangle = \underbrace{\left( \frac{\langle Q_P(o_0), q_t \rangle}{a_0} - \langle o_0, q_t \rangle \right)}_{\text{static RaBitQ error}} + \underbrace{\langle o_0 - o_t, q_t \rangle}_{\text{drift bias}},$$

whose magnitude is bounded by the drift certificate  $\gamma_t$  (Theorem 9.5) and which crosses the static RaBitQ stochastic envelope at a critical scale  $\Delta_t^{\text{crit}} = \Theta(1/\sqrt{D})$  (Theorem 9.6). The within-cell channel is therefore operative under CT in a way it is not under FF.

Read together, the two analyses describe the same drift event from two endpoints of a deployment spectrum. The same bounded centroid displacement  $\|c_T - c_0\|_2 \leq \varepsilon$  produces qualitatively different deployed-system behavior depending on which pieces of the original geometry the system updates. Drift does not disappear; it is routed — by deployment policy — into one channel or another.

### 12.2 Asymmetric repair cost

The two channels into which the two policies route drift are not equally remediable.

The within-cell channel that CT activates admits an exact, low-cost repair. By Theorem 10.3, given any valid drift certificate  $\gamma_t \geq \|o_t - o_0\|_2$ , the only stored sign-code coordinates that can possibly have flipped are those whose old rotated magnitude is at most  $\gamma_t$ . Refreshing exactly this *certified update set* recovers the full current sign code with no error and without touching any other coordinate. Under a Haar-random rotation, the expected size of the certified update set is governed by the dimensionless drift parameter  $\Lambda_t = \gamma_t \sqrt{D}$ : by Theorem 11.4, the expected update fraction converges to  $2\Phi(\Lambda_t) - 1 \approx \sqrt{2/\pi} \Lambda_t$  in the small-drift regime. The repair cost per affected vector is therefore  $O(\Lambda_t D)$  in expectation: linear in the dimension, proportional to the drift magnitude, and independent of the dataset size  $N_0 + T$ .

The inter-cell channel that FF activates does not admit an analogous repair. A full re-assignment requires recomputing  $\arg \min_k \|z_i - c_k^T\|_2^2$  for every stored vector  $z_i$ , an  $O(N_0 + T)$  operation. The natural question is whether some certificate-style shortcut — analogous to the margin condition  $|(y_0)_j| > \gamma_t$  that certifies a sign-code coordinate as stable — could exclude points whose assignment cannot have flipped using only build-time information. The structural obstacle is that the sign-code certificate exploits a continuity property: the rotated coordinates  $y_t = P^\top o_t$  are an isometry of the normalized directions, so a bound on  $\|o_t - o_0\|_2$  transfers coordinate-wise through the orthogonality of  $P$ . Cell assignment has no analogous structure. The map  $z \mapsto \arg \min_k \|z - c_k\|_2^2$  is a discrete decision over  $K$  alternatives whose value can flip between any two centroids whenever the realized drift moves  $z$  across a Voronoi boundary — a hyperplane whose location is itself a function of the drifted centroids. Without inspecting  $z$  against the fresh centroids, no build-time quantity bounds whether  $z$ 's assignment is stable: the boundary-band analysis of Theorems 6.2 and 6.8 controls the *aggregate* mass of assignment-drift but does not localize it without per-point evaluation.

The asymmetry between the two channels' repair costs is therefore not incidental. It reflects the difference between a continuous quantity (sign-code margin) for which a Lipschitz-style transfer of drift bounds exists, and a discrete quantity (cell assignment) for which no such transfer is available.

### 12.3 Implication for the deployment choice

The conventional reading of FF and CT as competing policies treats the choice as a trade-off between maintenance cost and deployed accuracy. The unified analysis suggests a different reading. Both policies absorb the same drift event  $\|c_T - c_0\|_2 \leq \varepsilon$ ; they differ in which channel absorbs it. FF concentrates drift in the inter-cell channel, where exact repair is  $O(N_0 + T)$ ; CT concentrates drift in the within-cell channel, where exact repair is  $O(\Lambda_t D)$  per affected point. Within the regime in which CT is well-defined — multi-centroid layouts where cell-level routing is meaningful — CT therefore routes drift toward the channel that admits inexpensive certified repair.

The design principle this suggests is not that CT is uniformly better. The within-cell penalty CT incurs is real: the deterministic drift term in Theorem 9.1 is non-zero whenever the drift is non-trivially aligned with the query. The argument is rather that, when refresh is on the table at all, CT places the system in a configuration where refresh can be performed exactly and locally, whereas FF places it in a configuration where refresh requires either full re-quantization or a full re-assignment scan. The choice between the two policies is therefore not only about which deployed accuracy profile is preferable; it is also about which drift channel one is willing to commit to repairing, and at what cost.

### 12.4 Limitations and outlook

Several aspects of the analysis are deliberately narrow and warrant explicit acknowledgment.

**Single-shot drift.** Both Part I and Part II analyze a single endpoint  $c_0 \rightarrow c_T$  with bounded displacement  $\|c_T - c_0\|_2 \leq \varepsilon$ . We do not model the trajectory along which the centroid drifts, nor the dynamics of repeated refresh under continuing drift. A streaming extension would need to model how the drift certificate  $\gamma_t$  evolves between refresh events and how repeated certified partial refreshes interact with continuing centroid evolution.

**Lower-bound formalization for inter-cell repair.** The argument that FF's inter-cell repair admits no certificate-style shortcut is structural rather than a formal lower bound. A precise statement would require fixing a class of admissible repair operations — those that use only

build-time information — and proving that any such operation must inspect a constant fraction of the dataset to certify assignment stability for boundary-band points. We state the obstruction at the level of mechanism (continuity vs. discreteness) rather than as a theorem.

**Margin storage assumption.** Theorem 10.3 assumes access to the rotated coordinates  $y_0 = P^\top o_0$ , or at least to lower bounds on  $|y_0|$ , in order to construct the certified update set. A standard RaBitQ deployment stores only the sign code and the alignment scalar [7]; the index would need to be augmented with per-coordinate margin information (or quantized lower bounds thereof) to support certified partial refresh. Whether to store exact rotated coordinates, quantized margins, or coarser blockwise certificates is a design choice that interacts with the storage budget and is not analyzed here.

**Hybrid policies.** We have considered the two endpoints of the deployment spectrum. Hybrid policies — for example, periodic batch refresh of the routing centroids combined with on-demand certified partial refresh of stored codes, or threshold-triggered policies driven by  $\Phi_T$  and the boundary-band masses — occupy the interior of this spectrum. The unified framing developed here is intended to support such analyses but does not itself carry them out.

**Assignment refresh under CT.** CT keeps routing geometry fresh by tracking centroids, but the *cell membership* of stored points is itself stale: a point  $z_i$  assigned to cell  $k$  at build time may, under fresh geometry, belong in a different cell. Theorem 10.3 refreshes the sign code of  $z_i$  in its current cell, not its cell assignment. Reconciling stored cell membership with fresh assignment is exactly the inter-cell repair problem analyzed for FF; it remains  $O(N_0 + T)$  in the worst case under CT as well. CT defers this problem rather than eliminating it.

The merger of the two parts of this report does not produce a single combined theorem; it produces a framing in which the two parts’ results are read as complementary specializations of a common question: given a frozen RaBitQ index and a bounded centroid drift, where does the drift end up, and what does it cost to remove? The asymmetry between the two channels’ repair costs answers the second half of that question, and supplies a structural reason why the deployment-policy choice matters.

## A Refinement of the leading constant in Theorem 4.1

Theorem 4.1 bounds  $\|\Delta \mathbf{o}\|_2 \leq 2\varepsilon/\rho$  via a uniform argument valid across the entire regularity domain. A more refined angular analysis in the two-dimensional plane spanned by  $\mathbf{u} := \mathbf{o}_r - \mathbf{c}_0$  and  $\boldsymbol{\delta} := \mathbf{c}_T - \mathbf{c}_0$  yields a leading constant of 1 rather than 2 (and analogously for the query side). We retain the constant-2 form in the body for its uniform validity over the regularity domain; the qualitative threshold analysis of Section 5 is insensitive to this factor.

## B Tightness of the union bound in Theorem 4.6

The two stochastic terms  $(A_0)$  and  $(A_T)$  in the proof of Theorem 4.6 share the same Haar matrix  $\mathbf{P}$ , and their input pairs differ in  $\ell_2$  by  $O(\varepsilon/\rho + \varepsilon/\sigma_0)$ . The union step ignores this coupling. A coupled analysis would require either a margin condition excluding coordinates of  $\mathbf{P}^{-1}\mathbf{o}$  near zero (since the bit-code map  $\mathbf{1}[\mathbf{P}^{-1}\mathbf{o} \geq \mathbf{0}]$  is discontinuous in  $\mathbf{P}$ ), or a high-probability Lipschitz statement averaged over  $\mathbf{P}$ . We leave such an analysis to future work; the union form suffices for the qualitative regime partition because Theorem 4.4 establishes that the deployed system’s pointwise retrieval accuracy does not depend on the tightness of  $\mathcal{E}_{\text{quant}}$ .

## C Multi-centroid bridge sketch

For  $K > 2$  centroids, the assignment  $a_0(\mathbf{x}) = \min \arg \min_{k \in [K]} \|\mathbf{x} - \mathbf{c}_k^0\|_2$  partitions  $\mathbb{R}^D$  into a Voronoi diagram with up to  $\binom{K}{2}$  pairwise hyperplanes  $H_0^{(j,k)}$ . An assignment change  $a_0(\mathbf{x}) = j \neq k = a_T(\mathbf{x})$  requires that the *pairwise* margin  $g_{jk}^t(\mathbf{x}) := \|\mathbf{x} - \mathbf{c}_j^t\|_2^2 - \|\mathbf{x} - \mathbf{c}_k^t\|_2^2$  change sign between  $t = 0$  and  $t = T$ ; the same Theorem 6.1 + Theorem 6.3 argument applies to that pair, and a union bound over the  $\binom{K}{2}$  pairs yields the candidate aggregate bound

$$\widehat{\mathcal{D}}_T(A_\Delta) \lesssim \sum_{1 \leq j < k \leq K} M_{\widehat{\mathcal{D}}_T}(H_0^{(j,k)}, \eta^{(j,k)}(\varepsilon)), \quad \eta^{(j,k)}(\varepsilon) := \frac{2R\varepsilon + \varepsilon^2/2}{\|\mathbf{c}_k^0 - \mathbf{c}_j^0\|_2}.$$

We present this as a *bridge sketch*, not a formal theorem. A rigorous  $K$ -centroid statement requires fixing additional conventions:

1. deterministic tie-breaking across all  $K$  clusters,
2. pairwise separation  $\|\mathbf{c}_j^0 - \mathbf{c}_k^0\|_2 > 0$  for the relevant pairs,
3. per-pair uniform radii  $R^{(j,k)}$ ,
4. whether  $\varepsilon$  is shared across centroids or centroid-specific,
5. whether the bound restricts to *adjacent* Voronoi neighbors (which can sharpen the union bound) or runs over all pairs.

The dual-centroid case of Section 6 is the minimal setting that exposes the mechanism; the  $K$ -centroid generalization is mechanically straightforward but conditionally heavy.

## D Proofs for the Problem Setup

### D.1 Proof of Lemma 8.1

*Proof.* Let  $v \in P\mathcal{C}$ . Then there exists  $x \in \mathcal{C}$  such that  $v = Px$ . Since  $P$  is orthogonal and  $\|o\|_2 = \|x\|_2 = 1$ , minimizing  $\|o - Px\|_2$  is equivalent to minimizing its square:

$$\|o - Px\|_2^2 = \|o\|_2^2 + \|Px\|_2^2 - 2\langle o, Px \rangle = 2 - 2\langle P^\top o, x \rangle.$$

Thus the nearest-codeword problem is equivalent to maximizing

$$\langle P^\top o, x \rangle$$

over  $x \in \mathcal{C}$ . Write  $y = P^\top o$ . Since each coordinate of  $x \in \mathcal{C}$  satisfies

$$x_j \in \left\{ -\frac{1}{\sqrt{D}}, +\frac{1}{\sqrt{D}} \right\},$$

we have

$$\langle y, x \rangle = \sum_{j=1}^D y_j x_j.$$

This objective separates coordinatewise. For each  $j$ , the term  $y_j x_j$  is maximized by choosing

$$x_j = \frac{\text{sign}_+(y_j)}{\sqrt{D}},$$

with the convention  $\text{sign}_+(0) = +1$ . Therefore the maximizing codeword is

$$x = \frac{\sigma}{\sqrt{D}}, \quad \sigma_j = \text{sign}_+(y_j).$$

The corresponding point in the rotated codebook is

$$Px = P \frac{\sigma}{\sqrt{D}} = Q_P(o).$$

Hence  $Q_P(o)$  is a nearest codeword in  $PC$ . □

## D.2 Proof of the Centered Distance Identity

*Proof.* Let

$$o = N_c(x) = \frac{x - c}{\|x - c\|_2}, \quad q = N_c(y) = \frac{y - c}{\|y - c\|_2},$$

and define

$$\alpha = \|x - c\|_2, \quad \beta = \|y - c\|_2.$$

Then

$$x - c = \alpha o, \quad y - c = \beta q.$$

Subtracting gives

$$x - y = (x - c) - (y - c) = \alpha o - \beta q.$$

Therefore,

$$\begin{aligned} \|x - y\|_2^2 &= \|\alpha o - \beta q\|_2^2 \\ &= \alpha^2 \|o\|_2^2 + \beta^2 \|q\|_2^2 - 2\alpha\beta \langle o, q \rangle. \end{aligned}$$

Since  $o, q \in \mathbb{S}^{D-1}$ , we have  $\|o\|_2 = \|q\|_2 = 1$ . Hence

$$\|x - y\|_2^2 = \alpha^2 + \beta^2 - 2\alpha\beta \langle o, q \rangle.$$

This proves the identity. □

### D.3 Proof of Proposition 8.5

*Proof.* Let

$$u := z - c_0, \quad v := z - c_t.$$

Then

$$N_{c_0}(z) = \frac{u}{\|u\|_2}, \quad N_{c_t}(z) = \frac{v}{\|v\|_2}.$$

Also,

$$u - v = (z - c_0) - (z - c_t) = c_t - c_0,$$

so

$$\|u - v\|_2 = \|c_t - c_0\|_2.$$

We want to bound

$$\left\| \frac{u}{\|u\|_2} - \frac{v}{\|v\|_2} \right\|_2.$$

Add and subtract  $v/\|u\|_2$ :

$$\begin{aligned} \left\| \frac{u}{\|u\|_2} - \frac{v}{\|v\|_2} \right\|_2 &\leq \left\| \frac{u}{\|u\|_2} - \frac{v}{\|u\|_2} \right\|_2 + \left\| \frac{v}{\|u\|_2} - \frac{v}{\|v\|_2} \right\|_2 \\ &= \frac{\|u - v\|_2}{\|u\|_2} + \|v\|_2 \left| \frac{1}{\|u\|_2} - \frac{1}{\|v\|_2} \right|. \end{aligned}$$

For the second term,

$$\begin{aligned} \|v\|_2 \left| \frac{1}{\|u\|_2} - \frac{1}{\|v\|_2} \right| &= \|v\|_2 \frac{|\|v\|_2 - \|u\|_2|}{\|u\|_2 \|v\|_2} \\ &= \frac{|\|v\|_2 - \|u\|_2|}{\|u\|_2}. \end{aligned}$$

By the reverse triangle inequality,

$$|\|v\|_2 - \|u\|_2| \leq \|v - u\|_2.$$

Therefore,

$$\begin{aligned} \left\| \frac{u}{\|u\|_2} - \frac{v}{\|v\|_2} \right\|_2 &\leq \frac{\|u - v\|_2}{\|u\|_2} + \frac{\|u - v\|_2}{\|u\|_2} \\ &= \frac{2\|u - v\|_2}{\|u\|_2}. \end{aligned}$$

By Assumption 8.4,

$$\|u\|_2 = \|z - c_0\|_2 \geq r_{\min}.$$

Thus,

$$\|N_{c_0}(z) - N_{c_t}(z)\|_2 \leq \frac{2\|u - v\|_2}{r_{\min}} = \frac{2\|c_t - c_0\|_2}{r_{\min}}.$$

This proves the claimed bound. Since  $\Delta_t = \|c_t - c_0\|_2$ , it follows that

$$\gamma_t := \frac{2\Delta_t}{r_{\min}}$$

is a valid drift certificate. □

## E Proofs for the Error Decomposition

This section contains the proofs for the decomposition and bias statements in Section 9. These arguments are deterministic algebraic consequences of the definitions, except for Corollary 9.2, which additionally invokes the static RaBitQ unbiasedness guarantee for the fixed pair  $(o_0, q_t)$ .

### E.1 Proof of Proposition 9.1

*Proof.* By definition of the stale estimator,

$$\hat{s}_{P,t}^{\text{stale}} = \frac{\langle Q_P(o_0), q_t \rangle}{a_0}, \quad a_0 = \langle Q_P(o_0), o_0 \rangle.$$

Subtracting the current target gives

$$\hat{s}_{P,t}^{\text{stale}} - \langle o_t, q_t \rangle = \frac{\langle Q_P(o_0), q_t \rangle}{a_0} - \langle o_t, q_t \rangle.$$

Add and subtract  $\langle o_0, q_t \rangle$ :

$$\begin{aligned} \hat{s}_{P,t}^{\text{stale}} - \langle o_t, q_t \rangle &= \frac{\langle Q_P(o_0), q_t \rangle}{a_0} - \langle o_0, q_t \rangle + \langle o_0, q_t \rangle - \langle o_t, q_t \rangle \\ &= \left( \frac{\langle Q_P(o_0), q_t \rangle}{a_0} - \langle o_0, q_t \rangle \right) + \langle o_0 - o_t, q_t \rangle. \end{aligned}$$

This is the desired decomposition. □

### E.2 Proof of Corollary 9.2

*Proof.* By the static RaBitQ unbiasedness guarantee applied to the fixed pair  $(o_0, q_t)$ ,

$$\mathbb{E}_P \left[ \frac{\langle Q_P(o_0), q_t \rangle}{\langle Q_P(o_0), o_0 \rangle} \right] = \langle o_0, q_t \rangle.$$

Since

$$a_0 = \langle Q_P(o_0), o_0 \rangle,$$

the stale estimator satisfies

$$\mathbb{E}_P [\hat{s}_{P,t}^{\text{stale}}] = \langle o_0, q_t \rangle.$$

Subtracting the current target  $\langle o_t, q_t \rangle$ , we obtain

$$\begin{aligned} \mathbb{E}_P [\hat{s}_{P,t}^{\text{stale}}] - \langle o_t, q_t \rangle &= \langle o_0, q_t \rangle - \langle o_t, q_t \rangle \\ &= \langle o_0 - o_t, q_t \rangle. \end{aligned}$$

This proves the claimed bias identity. □

### E.3 Proof of Lemma 9.4

*Proof.* By the Cauchy–Schwarz inequality,

$$|\langle o_0 - o_t, q_t \rangle| \leq \|o_0 - o_t\|_2 \|q_t\|_2.$$

Since  $q_t \in \mathbb{S}^{D-1}$ , we have

$$\|q_t\|_2 = 1.$$

Therefore,

$$|\langle o_0 - o_t, q_t \rangle| \leq \|o_0 - o_t\|_2.$$

□

## E.4 Proof of Corollary 9.5

*Proof.* By Proposition 9.1,

$$\hat{s}_{P,t}^{\text{stale}} - \langle o_t, q_t \rangle = \left( \frac{\langle Q_P(o_0), q_t \rangle}{a_0} - \langle o_0, q_t \rangle \right) + \langle o_0 - o_t, q_t \rangle.$$

Taking absolute values and applying the triangle inequality gives

$$|\hat{s}_{P,t}^{\text{stale}} - \langle o_t, q_t \rangle| \leq \left| \frac{\langle Q_P(o_0), q_t \rangle}{a_0} - \langle o_0, q_t \rangle \right| + |\langle o_0 - o_t, q_t \rangle|.$$

By Lemma 9.4,

$$|\langle o_0 - o_t, q_t \rangle| \leq \|o_0 - o_t\|_2.$$

Substituting this bound yields

$$|\hat{s}_{P,t}^{\text{stale}} - \langle o_t, q_t \rangle| \leq \left| \frac{\langle Q_P(o_0), q_t \rangle}{a_0} - \langle o_0, q_t \rangle \right| + \|o_0 - o_t\|_2.$$

□

## F Proofs for Certified Partial Refresh

This section proves the deterministic sign-stability results used in Section 10. The arguments use only the orthogonality of  $P$  and the drift certificate  $\|o_t - o_0\|_2 \leq \gamma_t$ .

### F.1 Proof of Lemma 10.2

*Proof.* Let

$$y_0 := P^\top o_0, \quad y_t := P^\top o_t.$$

Since  $P$  is orthogonal,  $P^\top$  preserves Euclidean norms. Therefore,

$$\|y_t - y_0\|_2 = \|P^\top o_t - P^\top o_0\|_2 = \|P^\top(o_t - o_0)\|_2 = \|o_t - o_0\|_2.$$

By the assumed drift certificate,

$$\|o_t - o_0\|_2 \leq \gamma_t.$$

Thus,

$$\|y_t - y_0\|_2 \leq \gamma_t.$$

For any coordinate  $j$ ,

$$|(y_t)_j - (y_0)_j| \leq \|y_t - y_0\|_2 \leq \gamma_t.$$

Now suppose

$$|(y_0)_j| > \gamma_t.$$

Then the perturbation from  $(y_0)_j$  to  $(y_t)_j$  is strictly smaller than the distance from  $(y_0)_j$  to zero. Hence the coordinate cannot cross zero. Therefore,

$$\text{sign}_+((y_t)_j) = \text{sign}_+((y_0)_j).$$

□



## F.2 Proof of Theorem 10.3

*Proof.* Let

$$\sigma_0 := \text{sign}_+(P^\top o_0), \quad \sigma_t := \text{sign}_+(P^\top o_t),$$

and let  $\tilde{\sigma}$  be the partially refreshed sign vector defined by

$$\tilde{\sigma}_j = \begin{cases} \sigma_{t,j}, & j \in \mathcal{U}_t, \\ \sigma_{0,j}, & j \notin \mathcal{U}_t. \end{cases}$$

We prove that  $\tilde{\sigma}_j = \sigma_{t,j}$  for every coordinate  $j$ .

If  $j \in \mathcal{U}_t$ , then this holds by definition, since the update rule explicitly sets

$$\tilde{\sigma}_j = \sigma_{t,j}.$$

Now suppose  $j \notin \mathcal{U}_t$ . By definition of  $\mathcal{U}_t$ ,

$$|(P^\top o_0)_j| > \gamma_t.$$

Since the theorem assumes

$$\|o_t - o_0\|_2 \leq \gamma_t,$$

Lemma 10.2 implies

$$\text{sign}_+((P^\top o_t)_j) = \text{sign}_+((P^\top o_0)_j).$$

That is,

$$\sigma_{t,j} = \sigma_{0,j}.$$

But outside  $\mathcal{U}_t$ , the update rule keeps the old sign:

$$\tilde{\sigma}_j = \sigma_{0,j}.$$

Therefore,

$$\tilde{\sigma}_j = \sigma_{t,j}.$$

Thus  $\tilde{\sigma}_j = \sigma_{t,j}$  for every coordinate  $j$ , so

$$\tilde{\sigma} = \sigma_t.$$

Finally, by the definition of the RaBitQ quantized vector,

$$Q_P(o_t) = P \frac{\sigma_t}{\sqrt{D}}.$$

Since  $\tilde{\sigma} = \sigma_t$ , we obtain

$$P \frac{\tilde{\sigma}}{\sqrt{D}} = P \frac{\sigma_t}{\sqrt{D}} = Q_P(o_t).$$

□

## G Random-Rotation Margin Facts

This section records the random-rotation facts used in Section 11. The certified refresh theorem itself is deterministic, but the scaling discussion assumes that the RaBitQ rotation is Haar-random.

## G.1 Haar Rotation Sends a Fixed Vector to a Uniform Sphere Vector

**Proposition G.1** (Haar rotation of a fixed vector). *Let  $P \sim \text{Haar}(O(D))$ , and let  $o_0 \in \mathbb{S}^{D-1}$  be fixed. Then*

$$P^\top o_0 \sim \text{Unif}(\mathbb{S}^{D-1}).$$

*Proof.* Let

$$Y := P^\top o_0.$$

Since  $P$  is orthogonal and  $o_0 \in \mathbb{S}^{D-1}$ , we have  $Y \in \mathbb{S}^{D-1}$ .

To show that  $Y$  is uniformly distributed on the sphere, it suffices to show that its distribution is invariant under every orthogonal transformation. Let  $R \in O(D)$  be fixed. Then

$$RY = RP^\top o_0.$$

Since  $P \sim \text{Haar}(O(D))$ , the transpose  $P^\top$  is also Haar-distributed on  $O(D)$ . Moreover, left multiplication by a fixed orthogonal matrix preserves Haar measure, so

$$RP^\top \sim P^\top.$$

Therefore,

$$RY = RP^\top o_0 \stackrel{d}{=} P^\top o_0 = Y.$$

Thus the law of  $Y$  is invariant under all rotations and reflections in  $O(D)$ . The unique probability measure on  $\mathbb{S}^{D-1}$  with this invariance property is the uniform surface measure. Hence

$$P^\top o_0 \sim \text{Unif}(\mathbb{S}^{D-1}).$$

□

## G.2 Proof of Proposition 11.1

*Proof.* Let  $X \sim \text{Unif}(\mathbb{S}^{D-1})$ . By symmetry, the density of  $X_1$  depends only on the size of the slice of the sphere with first coordinate equal to  $u$ .

Fix  $u \in [-1, 1]$ . The set of points  $x \in \mathbb{S}^{D-1}$  satisfying  $x_1 = u$  is a  $(D-2)$ -dimensional sphere in the remaining  $D-1$  coordinates:

$$x_2^2 + \cdots + x_D^2 = 1 - u^2.$$

Thus its radius is

$$\sqrt{1 - u^2}.$$

The surface area of a  $(D-2)$ -sphere of radius  $r$  scales as

$$r^{D-2}.$$

However, when converting spherical surface measure into a density with respect to  $du$ , there is an additional Jacobian factor

$$\frac{1}{\sqrt{1 - u^2}}.$$

Therefore the marginal density is proportional to

$$(1 - u^2)^{(D-2)/2} \cdot (1 - u^2)^{-1/2} = (1 - u^2)^{(D-3)/2}.$$

Hence

$$p_D(u) = C_D (1 - u^2)^{(D-3)/2}, \quad u \in [-1, 1],$$

for a normalizing constant  $C_D$ .

It remains to compute  $C_D$ . We require

$$1 = \int_{-1}^1 C_D (1 - u^2)^{(D-3)/2} du.$$

Using the beta-function identity,

$$\int_{-1}^1 (1 - u^2)^{(D-3)/2} du = B\left(\frac{1}{2}, \frac{D-1}{2}\right) = \frac{\Gamma(1/2)\Gamma((D-1)/2)}{\Gamma(D/2)}.$$

Since  $\Gamma(1/2) = \sqrt{\pi}$ , we obtain

$$C_D = \frac{\Gamma(D/2)}{\sqrt{\pi}\Gamma((D-1)/2)}.$$

Therefore,

$$p_D(u) = \frac{\Gamma(D/2)}{\sqrt{\pi}\Gamma((D-1)/2)} (1 - u^2)^{(D-3)/2}, \quad u \in [-1, 1].$$

Finally, since the density is symmetric about zero, for  $0 \leq \gamma \leq 1$ ,

$$\mathbb{P}(|X_1| \leq \gamma) = 2 \int_0^\gamma \frac{\Gamma(D/2)}{\sqrt{\pi}\Gamma((D-1)/2)} (1 - u^2)^{(D-3)/2} du.$$

□

### G.3 Proof of Proposition 11.3

*Proof.* Let

$$y_0 := P^\top o_0.$$

By Proposition G.1,

$$y_0 \sim \text{Unif}(\mathbb{S}^{D-1}).$$

The certified update set is

$$\mathcal{U}_t = \{j : |(y_0)_j| \leq \gamma_t\}.$$

For each coordinate  $j$ , define the indicator random variable

$$I_j := \mathbf{1}\{|(y_0)_j| \leq \gamma_t\}.$$

Then

$$|\mathcal{U}_t| = \sum_{j=1}^D I_j,$$

and therefore

$$\frac{|\mathcal{U}_t|}{D} = \frac{1}{D} \sum_{j=1}^D I_j.$$

Taking expectations and using linearity of expectation,

$$\begin{aligned} \mathbb{E}\left[\frac{|\mathcal{U}_t|}{D}\right] &= \frac{1}{D} \sum_{j=1}^D \mathbb{E}[I_j] \\ &= \frac{1}{D} \sum_{j=1}^D \mathbb{P}(|(y_0)_j| \leq \gamma_t). \end{aligned}$$

Since  $y_0$  is uniformly distributed on the sphere, all coordinates have the same marginal distribution. Thus,

$$\mathbb{P}(|(y_0)_j| \leq \gamma_t) = \mathbb{P}(|X_1| \leq \gamma_t), \quad X \sim \text{Unif}(\mathbb{S}^{D-1}).$$

Because  $|X_1| \leq 1$  always, this is equivalently

$$\mathbb{P}(|X_1| \leq \min\{\gamma_t, 1\}).$$

Therefore,

$$\mathbb{E}\left[\frac{|\mathcal{U}_t|}{D}\right] = \mathbb{P}(|X_1| \leq \min\{\gamma_t, 1\}).$$

If  $0 \leq \gamma_t \leq 1$ , applying Proposition 11.1 gives

$$\mathbb{P}(|X_1| \leq \gamma_t) = 2 \int_0^{\gamma_t} \frac{\Gamma(D/2)}{\sqrt{\pi} \Gamma((D-1)/2)} (1-u^2)^{(D-3)/2} du.$$

If  $\gamma_t \geq 1$ , then every coordinate satisfies

$$|(y_0)_j| \leq 1 \leq \gamma_t,$$

so  $\mathcal{U}_t = \{1, \dots, D\}$  and the update fraction is 1. This proves the claim.

No independence between coordinates is required; the argument uses only linearity of expectation and symmetry of the coordinate marginals.  $\square$

#### G.4 Small- $\Lambda$ Approximation

The exact expected update fraction is

$$\mathbb{E}\left[\frac{|\mathcal{U}_t|}{D}\right] = \mathbb{P}(|X_1| \leq \gamma_t), \quad X \sim \text{Unif}(\mathbb{S}^{D-1}).$$

Setting

$$\gamma_t = \frac{\Lambda_t}{\sqrt{D}},$$

this becomes

$$\mathbb{E}\left[\frac{|\mathcal{U}_t|}{D}\right] = \mathbb{P}(\sqrt{D} |X_1| \leq \Lambda_t).$$

A standard high-dimensional approximation is that

$$\sqrt{D} X_1 \Rightarrow Z, \quad Z \sim \mathcal{N}(0, 1),$$

as  $D \rightarrow \infty$ , where  $\Rightarrow$  denotes convergence in distribution. Under this approximation,

$$\mathbb{P}(\sqrt{D} |X_1| \leq \Lambda_t) \approx \mathbb{P}(|Z| \leq \Lambda_t).$$

Thus,

$$\mathbb{E}\left[\frac{|\mathcal{U}_t|}{D}\right] \approx 2\Phi(\Lambda_t) - 1,$$

where  $\Phi$  is the standard normal cumulative distribution function.

For small  $\Lambda_t$ , using the Taylor expansion of the Gaussian density near zero,

$$2\Phi(\Lambda_t) - 1 = \sqrt{\frac{2}{\pi}} \Lambda_t + O(\Lambda_t^3).$$

Therefore, in the small- $\Lambda_t$  regime,

$$\mathbb{E}\left[\frac{|\mathcal{U}_t|}{D}\right] \approx \sqrt{\frac{2}{\pi}} \Lambda_t.$$

This approximation is not needed for the deterministic correctness theorem. It is only used to interpret the scaling behavior of the certified update fraction. The exact finite-dimensional quantity remains the spherical integral in Proposition 11.3.

## References

- [1] Noga Alon and Bo'az Klartag. Optimal compression of approximate inner products and dimension reduction. In *Proceedings of the 58th IEEE Annual Symposium on Foundations of Computer Science*, pages 639–650, 2017.
- [2] Albert Bifet and Ricard Gavaldà. Learning from time-changing data with adaptive windowing. In *Proceedings of the 2007 SIAM International Conference on Data Mining*, pages 443–448, 2007.
- [3] Moses S. Charikar. Similarity estimation techniques from rounding algorithms. In *Proceedings of the 34th Annual ACM Symposium on Theory of Computing*, pages 380–388, 2002.
- [4] Mayur Datar, Nicole Immorlica, Piotr Indyk, and Vahab S. Mirrokni. Locality-sensitive hashing scheme based on p-stable distributions. In *Proceedings of the 20th Annual Symposium on Computational Geometry*, pages 253–262, 2004.
- [5] Matthijs Douze, Alexandr Guzhva, Chengqi Deng, Jeff Johnson, Gergely Szilvasy, Pierre-Emmanuel Mazaré, Maria Lomeli, Lucas Hosseini, and Hervé Jégou. The faiss library. *arXiv preprint arXiv:2401.08281*, 2024.
- [6] João Gama, Indrè Žliobaitė, Albert Bifet, Mykola Pechenizkiy, and Abdelhamid Bouchachia. A survey on concept drift adaptation. *ACM Computing Surveys*, 46(4):1–37, 2014.
- [7] Jianyang Gao and Cheng Long. RaBitQ: Quantizing high-dimensional vectors with a theoretical error bound for approximate nearest neighbor search. In *Proceedings of the ACM on Management of Data (SIGMOD)*, 2024.
- [8] Tiezheng Ge, Kaiming He, Qifa Ke, and Jian Sun. Optimized product quantization for approximate nearest neighbor search. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 2946–2953, 2013.
- [9] Ruiqi Guo, Philip Sun, Erik Lindgren, Quan Geng, David Simcha, Felix Chern, and Sanjiv Kumar. Accelerating large-scale inference with anisotropic vector quantization. In *Proceedings of the 37th International Conference on Machine Learning*, pages 3887–3896, 2020.
- [10] Hervé Jégou, Matthijs Douze, and Cordelia Schmid. Product quantization for nearest neighbor search. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 33(1): 117–128, 2011.
- [11] Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with gpus. *arXiv preprint arXiv:1702.08734*, 2017.
- [12] Aditi Singh, Suhas Jayaram Subramanya, Ravishankar Krishnaswamy, and Harsha Vardhan Simhadri. Freshdiskann: A fast and accurate graph-based ann index for streaming similarity search. *arXiv preprint arXiv:2105.09613*, 2021.
- [13] Suhas Jayaram Subramanya, Devvrit, Rohan Kadekodi, Ravishankar Krishnaswamy, and Harsha Vardhan Simhadri. Diskann: Fast accurate billion-point nearest neighbor search on a single node. In *Advances in Neural Information Processing Systems*, 2019.
- [14] Shay Vargaftik, Ran Ben-Basat, Amit Portnoy, Gal Mendelson, Yaniv Ben-Itzhak, and Michael Mitzenmacher. Drive: One-bit distributed mean estimation. In *Advances in Neural Information Processing Systems*, 2021.
- [15] Shay Vargaftik, Ran Ben Basat, Amit Portnoy, Gal Mendelson, Yaniv Ben Itzhak, and Michael Mitzenmacher. Eden: Communication-efficient and robust distributed mean

- estimation for federated learning. In *Proceedings of the 39th International Conference on Machine Learning*, pages 21984–22014, 2022.
- [16] Amir Zandieh, Majid Daliri, and Insu Han. Qjl: 1-bit quantized jl transform for kv cache quantization with zero overhead. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.
- [17] Amir Zandieh, Majid Daliri, Majid Hadian, and Vahab Mirrokni. Online vector quantization with near-optimal distortion rate. In *International Conference on Learning Representations*, 2026.